

VOLUME 1. BOOK 4

SECTION 10.3

GAS TURBINE POWER STATION (GTPS)

FEASIBILITY STUDY

**Construction of infrastructure facilities of special economic zone
“National industrial and petrochemical techno park” in Atyrau region**

Feasibility study composition

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10.3.1. GENERAL INFORMATION

10.3.3.1. BRIEF DESCRIPTION OF GAS TURBINE POWER STATION

Gas turbine power station with combined cycle is designed for production and supply of National industrial and petrochemical techno park in Atyrau region:

- High voltage power, 110 kV;
- Medium voltage power, 10.5 kV;
- High-pressure steam (exported);

GTPS is located within KPI field.

Four gas turbines with power 50 MW each, four exhaust heat boilers, two steam turbines with power 55 MW, 10/110 kV coupling transformer with a system and maintenance facilities are planned to be set up at GTPS.

Basic and maintenance facilities are primarily located in the main building of the power station. Exhaust stacks of exhaust heat boilers with height 70 m are adjusted outside A row side of the main building. Bypass exhaust stacks are set up along D columns, inside main building. Air condensers are equipped in an open area behind the 17 axis of main building. Diesel fuel and treated water tanks are located in an open area on A row side of the main building. Main transformers are adjusted outside on I row side. 35/10 kV transformers are located outside 1 axis. Production facilities are in between A-E rows and 1-7 axes.

Main technical and economic performance indexes

Ser. No.	Performance	Unit of measurements	Index
1	Installed capacity	MW	310
2	Parts of main facility: - gas turbine with generator – 4 sets - exhaust heat boiler – 4 sets - steam turbine with generator – 2 sets	MW t/h MW	50,0 (ISO Rating) 131,5 55,0
3	Used per year	h/y	8100
4	Type of fuel		Natural gas, ethane (back-up)
5	Annual fuel consumption, gas	tnt thous. thous. m ³	691,9 599 586,3
6	Total number of employees	persons	33
7	Engineering site area	m ²	49025
8	Land-to-building ratio	%	43,3
9	Construction area	sq.m	21240

10.3.2. MASTER PLAN AND TRANSPORT

Design concept on master plan and transport at gas turbine power station area is described in Volume 1, Book 2, Section 7.

10.3.3. MECHANICAL ANNEX

10.3.3.1. BRIEF DESCRIPTION OF GAS TURBINE POWER STATION

The gas turbine power station is designed to supply electrical power and steam for process purposes to integrated techno park located in Atyrau region of the Republic of Kazakhstan.

According to operation schedule there will be installed four gas turbines, four exhaust heat boilers, two steam turbines, and maintenance facility in order to produce electric power and steam. One gas turbine and exhaust heat boiler are spare and might be used during maximum load.

Gas turbine power plant is designed for operating utilizing two types of fuel:

- natural gas (main fuel)
- ethane (backup fuel)

The electric power produced at gas turbine power station mainly covers techno park needs. Part of electric power may be distributed to external power network.

10.3.3.1.1 TECHNOLOGICAL SYSTEM OF GAS TURBINE ELECTRIC POWER STATION

Technological system of gas turbine power station provides installation of equipment:

- four gas turbine power plants
- four exhaust heat boilers
- two steam turbines
- deaerator
- feed water pumps of exhaust heat boilers
- feed water pumps for operating procedure

Each exhaust heat boiler is followed by installation of:

- Continuous blow down expander
- Cyclic blow down expander
- Blow down water cooler
- Three pumps of blow down water

Each steam turbine is followed by installation of:

- Air condenser cooler
- Three condensate pump

10.3.3.2 EQUIPMENT OF GAS TURBINE ELECTRIC POWER STATION

10.3.3.2.1 GAS TURBINE POWER PLANT

Gas turbine power plant consists of following parts:

- Compressor;
- Combustor;
- Gas turbine;
- Reducer;
- Power source;

Multistage compressor in gas turbine electric power plant compresses the air and pushes it under high pressure to combustor. Air feed area of multistage compressor is fitted with an air filtration. Air and fuel go to combustor of gas turbine electric power plant at the same time. When fuel and air meet at high speed, they combust. The air-fuel mixture is combusted emitting large amount of energy. Then energy of combustion gas is converted into mechanical work by stream rotation of turbine blade incandescent gas. Part of produced energy is expended during compressing air in a compressor. The rest of operation is transmitted to electric generator.

Three-stage gas turbine of reactive type.

Electric power produced by generator is distributed to power consumers through transformer.

Exhaust gas goes to heat boiler for heat generation. After passing through heat boiler, the exhaust gas goes to exhaust stack and emits into the air.

When exhaust heat boiler stops, the exhaust gas goes to bypass exhaust stack and emits into the air.

10.3.3.2.1.1. COMPRESSOR OF GAS TURBINE POWER PLANT

Compressor of gas turbine electric power plant is designed for an air compression and its output to combustor. The fuel mixed with compressed air in combustor flames producing large volume of energy.

Compressor of gas turbine electric power plant rotates with the speed 5,100 r/min.

10.3.3.2.1.2. GAS TURBINE

The gas turbine is designed to convert energy of combustion gas into mechanical work by stream rotation of turbine blade incandescent gas. Flue gases in combustor expand and drive turbine reducing pressure and temperature at the turbine outlet. Three-stage gas turbine of reactive type.

Technical specification of gas turbine is shown in the Table 10.3.1.

Table 10.3.1 - Technical specification of gas turbine*

Ser. No.	Performance	Size	Value
1	Number of stages	-	3
2	Capacity	MW	50
3	Flow rate of combustion gas	m ³ /h	597.4
4	Temperature of combustion gas	°C	546.5
5	Output round	r/min	3,000
6	Flow rate of gas	m ³ /h	20,500
7	Flow rate of diesel fuel	m ³ /h	12.75

* Utilized equipment and its technical specification should be updated at designing last stages

10.3.3.2.1.3. GAS TURBINE LAUNCHING SYSTEM

The gas turbine is equipped with an automatic start system that has a starting power drive installed in the auxiliary mounting plate. Control panel of the gas turbine fitted with triggers signal button. Starting engine is connected to the gear with claw clutch transmitting starting torque to the shaft of a gas turbine. Hydraulic solenoid valve opens the claw clutch automatically when gas turbine reaches self-sustained speed of rotation.

10.3.3.2.1.4. ELECTRIC GENERATOR OF GAS TURBINE

Electric generator of an asynchronous type.

Technical specification of a generator is shown in the Table 10.3.2.

Table 10.3.2 - Technical specification of the gas turbine generator.

Ser. No.	Name	Size	Value
1	Electrical power	MV	50.4
2	Voltage	kV	10.5
3	Number of phases	-	3
4	Number of poles	-	2
5	Rotary speed	r/min	3000
6	Current frequency	Hz	50
7	Total weight	t	70.3
8	Rotary weight	t	20.0

10.3.3.2.2. EXHAUST HEAT BOILER

According to the operation schedule there following flows of the exhaust heat boiler:

- combustion gas input from gas turbine;
- feed water input;
- natural gas or propane-butane gas flow to exhaust reheating for maintaining rated value;
- air for gas combustion procedure;
- exhaust gas to exhaust stack;
- condensate in deaerator;
- fresh steam;
- continuous blow down;
- cyclic blow down.

Technical specification of the exhaust heat boiler is shown in the Table 10.3.3.

Table 10.3.3 - Technical specification of the exhaust heat boiler *

Ser. No.	Name	Size	Value
1	Steam capacity	t/h	131.5
2	Steam temperature	°C	530
3	Steam pressure	bar	70
4	Gas flow rate	m ³ /h	max. up to 12771

* Utilized equipment and its technical specification should be updated at designing last stages

Exhaust heat boiler is designed to generate steam by utilizing the heat of exhaust gases going through gas turbine.

Cycle efficiency of gas turbine plant increases when using heat of exhaust gases of the gas turbine. Exhaust heat boiler generates steam of high and low pressures.

Part of high-pressure steam is piped to the steam turbine for power generation, while the other part of steam is directed to production needs of the gas chemical complex.

Drum exhaust heat boiler with a natural circulation of evaporator circuits in high and low pressure, single-hull with suspension to its own structures through intermediate metal.

Following heating surfaces are sequentially arranged along gases of exhaust heat boiler:

- Screen;
- Final steam superheater;
- Secondary steam superheater;
- Primary steam superheater;
- Evaporator;
- Preheater;
- Water preheating equipment.

Each of two exhaust heat boiler systems consists of following parts:

- Steam preheating equipment;
- Preheater;
- Evaporator;
- Drum;

- Steam superheater;
- Boiler frame;
- Piping;
- Additional blowpipe.

10.3.3.2.2.1. CONDENSATE PREHEATING EQUIPMENT

Condensate preheating equipment is designed to utilize heat of exhaust gases going through gas turbine for primary condensate heating and further deliver to deaerator.

Pipes of heat change surface are mounted vertically and connected to upper and lower reservoirs with durable tight fusion, filled stands, racks, and connecting pipe at drum level.

10.3.3.2.2.2. PREHEATER

Preheater is made out of pipes with heat change surface area mounted vertically and connected to upper and lower collector with durable tight fusion.

Heated high-pressure condensate, discharged from preheater, goes to the evaporator.

10.3.3.2.2.3. EVAPORATOR

Evaporator has two sections:

- screen sections, which first cross exhaust gases;
- main evaporator.

Both sections have the form of finned tubes mounted vertically and connected to lower collectors and directly to drum with durable tight fusion

10.3.3.2.2.4. DRUM

The drum is taken out of flue gases and connected to the evaporator with waste and vertical pipes. The drum is made of carbon steel and consists of following parts:

- safety valve;
- overflow pipe;
- flow nipple to introduce chemical;
- flow nipple to introduce feed water;
- flow nipple for continues and cyclic blow down;
- pressure gauge;
- flow nipple for vapor sampling;
- flow nipple for water sampling;
- watermeter;
- manhole.

10.3.3.2.2.5. STEAM SUPERHEATERS

Steam superheater section is divided in two inter-stage desuperheaters:

- Primary steam superheater;
- Secondary steam superheater;
- Final steam superheater.

The primary steam superheater is first stage of the steam superheater system and generates low-temperature gas.

Superheated steam of primary superheater generates steam at intermediate level.

Steam goes from secondary steam superheater to final steam superheater.

All sections of the superheater are counter-current with vertical tubes welded to horizontal collectors.

Irrigation type desuperheaters of superheater sections are equipped with connecting pipes, and spray nozzles.

First desuperheater is located between the primary stem superheater, second desuperheater - between the secondary and final superheaters.

10.3.3.2.2.6. EXHAUST HEAT BOILER FRAME

Exhaust heat boiler frame and external casing are made of carbon steel. The outer casing is ribbed, gas-tight, and covered with thermal insulation.

10.3.3.2.2.7. ADDITIONAL BLOWPIPE

Exhaust heat boiler is equipped with additional blowpipe and forced-draught fan.

The forced-draught fan blows air to the blowpipe when the gas turbine is out of service.

10.3.3.2.3 STEAM TURBINE

Steam turbine is a heat engine in which steam energy is converted into mechanical work. In the steam turbine blade row the potential energy of water vapor is converted to kinetic energy, which in turn, is converted into mechanical rotation of the turbine shaft.

Steam turbine and generator comprise turbine generator.

The steam turbine body consists of fixed blades and a rotor with mounted rotating blades. The rotor is moved by steam passing through the blades. Steam goes from exhaust heat boiler to steam turbine.

10.3.3.2.3.1 PROCESS DESIGN OF STEAM TURBINE

Following flows are provided in steam turbine process design:

- exhaust heat boiler steam 1 - 4
- cooling condensate for exhaust heat boiler 1 - 4
- steam bypass of exhaust heat boiler 1 - 4
- turbine steam going to air condenser
- steam of exhaust heat boiler 1,2,3,4 for production.

Technical specification of the steam turbine is shown in the Table 10.3.4.

Table 10.3.4 - Technical specification of the steam turbine

Ser. No.	Name	Size	Value
1	Capacity	MW	55
2	Flow rate of fresh steam	t/h	263
3	Pressure of fresh steam	bar	70
4	Temperature of fresh steam	°C	530
5	Output steam pressure	bar	0.08
6	Output steam temperature	°C	41.5

10.3.3.2.3. 2. Electrical generator of steam turbine

Electric generator of an asynchronous type

Technical specification of the steam turbine generator is shown in the Table 10.3.5.

Table 10.3.5 - Technical specification of the steam turbine generator

Ser. No.	Name	Size	Value
1	Electrical power	MW	55
2	Voltage	kV	10.5

3	Number of phases	-	3
4	Number of poles	-	2
5	Rotary speed	r/min	3,000
6	Current frequency	Hz	50
7	Total weight	t	87
8	Rotary weight	t	15

10.3.3.3 FEED WATER SYSTEM

Feed water system consists of:

- deaerator;
- feeding pump;
- treated water tank
- pipelines

10.3.3.3.1 DEAERATOR

Deaerator is designed to remove corrosive gasses:

- out of feed water cycle;
- out of makeup treated water.

The deaerator consists of:

- deaerator tank;
- deaerator column.

Deaerator tank maintains:

- effective deaerating of treated water;
- sufficient store of treated water;
- stable operation of feeding pump.

Following flows are provided in deaerator process design:

- treated water to exhaust heat boilers;
- treated water to irrigate steam circuits;
- low-pressure steam of blow-down;
- turbine condensate;
- condensate of techno park production;
- condensate of exhaust heat boilers;
- makeup treated water;
- low-pressure heating steam.

The feed water is deaerated thermally in a deaerator column.

In order to deaerate feed water the low-pressure heating steam goes into deaeration column. Control valve is mounted on a steam pipeline to regulate the steam flow.

Evaporate from deaerator column goes to the air.

Deaerated feed water deaeration flows from the column into deaeration tank.

Feed water goes from deaerating tank to the suction feed pumps.

Technical specification of the deaerator is shown in the Table 10.3.6.

Table 10.3.6 - Technical specification of the deaerator

Ser. No.	Name	Size	Value
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1	Pressure	bar	5
2	Temperature of feed water	°C	220
3	Tank capacity	m ³	100

10.3.3.3.2 FEED WATER PUMPS

Feed water system has two groups of feed pumps:

- three feed pumps transporting deaerated water to exhaust heat boilers.

Two working pumps, one spare.

- three feed pumps transporting deaerated water for production needs.

Two working pumps, one spare.

Technical specification of feed pump of exhaust heat boiler is shown in the Table 10.3.7.

Table 10.3.7 - Technical specification of feed pump of exhaust heat boiler

Ser. No.	Name	Size	Value
1	Production rate	t/h	133
2	Pressure	m	1,114
3	Engine power	kW	710

Technical specification of feed pump for production needs is shown in the Table 10.3.8.

Table 10.3.8 - Technical specification of feed pump for production needs

Ser. No.	Name	Size	Value
1	Production rate	t/h	133
2	Pressure	m	745
3	Engine power	kW	710

10.3.3.3.3 STEAM-WATER CYCLE MAKEUP

There is provided a makeup of cycle treated water in order to replace loss of the steam cycle. Treated water is supplied from industrial complex to reserve tank of treated water with capacity of 2,000 m³. The treated water is pumped from the tank to the deaerator.

Technical specification of treated water pump is shown in the Table 10.3.9.

Table 10.3.9 - Technical specification of treated water pump

Ser. No.	Name	Size	Value
1	Production rate	t/h	98
2	Pressure	m	120
3	Engine power	kW	22

10.3.3.4. STEAM SUPPLY FACILITY

Steam supply facility is designed to supply steam for production and operating purposes.

Direct steam is supplied from high-pressure reservoir through reduction-cooling equipment to the following steam consumers:

- high-pressure steam for techno parks factories;
- low-pressure steam for techno parks factories;
- low-pressure steam for deaerator;
- low-pressure steam for HVAC room;
- low-pressure steam for depositor.

Reduction-cooling equipment consists of electrical valve, butterfly valve and steam cooling system.

Water for irrigation is supplied from the feed water pumps to the steam cooling system.

Motorized valve and control valve are fitted one by one on feed water irrigation pipe for steam coolers. Control valve regulates amount of water supplied to the steam cooling system.

10.3.3.5 SAMPLING SYSTEM

The sampling system of steam and water regulates the quality of water and steam.

If quality of water and steam do not conform with normative, they are sprayed with chemicals into appropriate paths to reach standard indicators.

Following substances should be checked:

- high-pressure saturated steam of exhaust heat boilers
- high-pressure superheated steam of exhaust heat boilers
- preheated condenser at deaerator input
- turbine condensate
- feed water
- blowdown water of exhaust heat boilers

10.3.3.6. AIR-COOLING SYSTEM OF CONDENSER

Process design provides following flows:

- steam going from steam turbine to condenser
- condensate going to steam consolidation condenser
- condensate to bypass desuperheater coil
- make-up water.

Mechanical-draft tower fans cool condensate in steam turbine condenser.

Technical specification of mechanical-draft tower is shown in the Table 10.3.10.

Table 10.3.10 - Technical specification of mechanical-draft tower

Ser. No.	Name	Size	Value
1	Number of fans	it	9
2	Pressure	m	745
3	Fan engine power	kW	132

Air condenser condenses the exhaust steam from steam turbine or cooled steam after superheating steam of turbine bypass with heat emission to the air.

Air condenser consists of finned tube beams.

Air condenser set consists of:

- steam pipe
- cooling air fans

- hot well
- three condensate pumps
- air condenser cleaning system,

Drainage system consists of following items:

- drainage tank
- expander
- two drainage pumps
- vacuum pump to remove air

10.3.3.6.1. EXHAUST STEAM REMOVAL FROM STEAM TURBINE

Exhaust steam goes from the last stage of a steam turbine to covered entry holes of finned tubes of air condenser. Steam goes in pipeline diameter of 3,400 mm towards air condenser.

10.3.3.6.2. CONDENSER OF STEAM TURBINE

Steam turbine condenser of air cooling system. Steam condenser is made of finned tubes tightened in beams and grouped into sections.

Each section has a fan. Fan runs air stream to cool the steam. Fans are equipped with variable speed drive to optimize air condenser, allowing selecting the optimum fan speed according to the heat load and environmental factors, and reducing energy consumption.

Air condenser is equipped with shutters preventing from icing in wintertime.

10.3.3.6.3 HOT WELL

Hot well is designed for collection condensate air condenser from finned tube beams and remainders returned to the drainage receptacle.

Soak period of hot well is 5 minutes.

Hot well also receives additional water from tank with treated water and recirculation line with a minimum consumption of condensate pumps.

10.3.3.6.4. CONDENSATE PUMPS

Condensate pumps direct condensate from hot well to the deaerator.

Condensate pumps also feed water to the steam turbine bypass system for cooling steam depressurized after overheating, before it goes into drainpipes, connected to the steam line, and water into steam turbine section with low-pressure to prevent hazardous heating.

Technical specification of condensate pump is described in 10.3.11.

Table 10.3.11 - Technical specification of condensate pump

Ser. No.	Name	Size	Value
1	Production rate	t/h	150
2	Pressure	m	120
3	Engine power	kW	100

10.3.3.6.5. DRAINAGE SYSTEM

Remains of the steam pipe go to drain pot of a drainage system.

Drainage pumps, located in catch pit under drainage pot, push the remains into hot well.

An expander, which collects condensed steam leakage of the steam turbine sealing system from seal gas condenser, expands condensate, where steam goes to the steam pipe for condensing by air condenser, while remains go to drainage pot.

10.3.3.6.6. DEAERATOR SYSTEM

Air and other incondensable substances should be removed from the condenser.

The vacuum system, designed to remove non-condensable gas, comprises two annular vacuum pumps. During startup both pumps achieve the vacuum operation, while during normal operation the vacuum is kept only by one pump.

10.3.3.7. CLOSED COOLING WATER SYSTEM

Closed cooling water system feeds cooled water to maintain equipment and devices.

Cooling water system of closed cycle is filled with condensed treated water.

Cooling water pumps of closed cycle feed cooling water to all gas turbine power station consumers. Water turning back by consumers is cooled in PNL cooling water system of closed cycle and pumped back by consumers' pumps.

Process design of closed cooling water system provides cooling water feed to the following cooling devices:

- electric generator of gas turbine 1 - 4
- oil cooler of gas turbine 1 - 4
- pump of cooling water
- pump of cooled condensate bend
- sample cooler
- HVAC room
- other consumers.

10.3.3.8. CONDENSATE SYSTEM

The condensate system provides following flows:

- condensate of discharge pumps in steam seal condenser
- steam from the turbine seal system
- condensate from the condenser steam seal in exhaust heat boiler 1 -4 for preheating

10.3.3.9. SEAL STEAM CONDENSER

Seal steam condenser is designed to condense steam from the turbine end seals. Seal steam condenser is a surface type heat exchanger. Steam, condensed in a condenser, flows by gravity into expansion tank.

10.3.3.10 BLOWDOWN SYSTEM OF EXHAUST HEAT BOILERS

Blowdown of exhaust heat boilers is performed to drain together with boiler water the hardness salts and excess sludge by simultaneous replacement of blown feed water that contains normal amount of salinity.

Blowdown of heat recovery boilers is performed continuously and periodically.

Continuous blowdown provides permanent removal of excess salts from boiler water during exhaust heat boilers operation. Continuous blowdown is performed from a boiler drum.

Boiler water from the drum goes to separator, in which water expands and steam separates. The steam goes from the separator to the feed water deaerator. Hot blow water then drains into the expander blowdown.

Intermittent blowdown of exhaust heat boilers is produced in a certain period of time and is intended mainly for removing sludge from the lower points of the boiler, drum and screen collectors.

Intermittent blowdown exhaust heat boilers is produced quickly, but with a large discharge of boiler water, which moves and carries sludge in drum or collectors and takes it to the intermittent blowdown separator.

Water expands and the steam separates in the intermittent blowdown separator.

Precooled blowdown water is pumped to the liquid waste from the separator.

Steam is discharged into the air from intermittent separator blowdown.

Intermittent blowdown of exhaust heat boilers is performed from lower points at each start and during scheduled operation of the boilers.

10.3.3.10.1 Continuous blowdown separator

Continuous blowdown separator is a vertical cylindrical vessel with flat or elliptical bottoms, steam and drainage pipes and float valve that automatically maintains water level in the separator.

Flow is twisted by tangential inlet of steam-water mixture to the inner wall of the separator. Separation of steam and water is produced in the middle part of the separator. Steam preserving rotational movement goes to steam space and is discharged through a nozzle disposed at the upper bottom. Water flows down the inner surface of the separator to the water zone and discharged through a nozzle at the frame lower part. There is a fitting at the lower bottom to drain water from separator when it is turned off and to clean lower water zone from sludge and dirt. Separator is mounted on legs to metal structures.

Technical specification of continuous blowdown separator is described in 10.3.12.

Table 10.3.12 - Technical specification of continuous blowdown separator

Ser. No.	Name	Size	Value
1	Volume	m ³	1.5
2	Pressure	bar	7

10.3.3.10.2 INTERMITTENT BLOWDOWN SEPARATOR

Intermittent blowdown separator is a vertical cylindrical vessel with elliptical bottoms

Flow is twisted by tangential inlet of steam-water mixture to the inner wall of the separator. Separation of steam and water is produced in the middle part of the separator. Steam preserving rotational movement goes to steam space and is discharged through a nozzle disposed at the upper bottom. Water flows down the inner surface of the separator to the water zone and discharged through a nozzle at the frame lower part. There is a fitting at the lower bottom to drain water from separator when it is turned off and to clean lower water zone from sludge and dirt. Separator is mounted on legs to metal structures.

Technical specification of intermittent blowdown separator is described in the Table 10.3.13.

Table 10.3.13 - Technical specification of intermittent blowdown separator

Ser. No.	Name	Size	Value
1	Volume	m ³	7.5
2	Pressure	bar	1.5

10.3.3.11. FUEL SUPPLY SYSTEM

10.3.3.11.1 GAS FUEL SYSTEM

Natural gas is a main fuel of a gas turbine plant.

High-pressure natural gas (37-39 bar) flows from outside to gas conditioning units.

Natural gas is fed into the gas turbine combustor from gas conditioning units.

Dry gas flows to the exhaust heat boiler through a pressure reduction station.

Pressure of natural gas after going through pressure reduction station is 5 bars.

In cases of interruption of natural gas flow, propane-butane gas can be used. Pressure of propane-butane is 5 bars.

Composition of natural gas is described in the Table 10.3.14.

Table 10.3.14 - Composition of natural gas

Ser. No.	Indicators	Requirements of standards of the RK	Actual
1	Low heat value by 20 °C, kcal/m ³	7,600	8,105
2	Wobbe index range, kcal/m ³	9,850-13,000	11,705
3	Weight concentration of hydrogen sulfide, g/m ³	0.007	0.0014
4	Weight concentration of sour sulfur, g/m ³	0.016	0.0021
5	Volume ratio of oxygen, %	1	0.006
6	Wight of mechanical impurities, g/m ³	0.001	
7	Composition	In volume, %	In molar, %
	CH ₄	94.609	94.573
	C ₂ H ₆	2.569	2.584
	C ₃ H ₈	0.473	0.48
	i-C ₄ H ₁₀	0.076	0.078
	n-C ₄ H ₁₀	0.101	0.105
	i-C ₅ H ₁₂	0.02	0.021
	n-C ₅ H ₁₂	0.022	0.023
	C ₆ H ₁₄	0.073	0.08
	Nitrogen	1.367	1.364
	Carbon dioxide	0.684	0.686
8	Water dew-point, C° by t _{gas} =37.2°C and P _{gas} =50.7 bars	4	
9	Water dew-point P _{gas} =40 bars	-6.6	
10	Density kg/m ³ , by 20 °C and 760 mm Hg (nominal)	0.71	
11	Density kg/m ³ , by 20 °C and 760 mm Hg (bottle method)	0.72	

10.3.3.11.2 DIESEL FUEL SYSTEM

Diesel fuel is a reserve fuel.

Technical specification of diesel fuel separator is described in 10.3.15.

Table 10.3.15 - Technical specification of diesel fuel

Ser. No.	Name	Size	Value
1	Net caloric value	kJ/kg	42,566

2	Density by 15°C	kg/l	0.8377
3	Viscosity by 40°C	cSt	4.15

Diesel fuel is delivered to the gas turbine power station by vehicles in tanks.

Diesel fuel is pumped by extraction pump into storage tank of diesel fuel with capacity of 1,000 m³ by tank pipeline through fuel filter.

Technical specification of diesel fuel extraction pump is described in the Table 10.3.16.

Table 10.3.16 - Technical specification of diesel fuel extraction pump

Ser. No.	Name	Size	Value
1	Production rate	t/h	30
2	Pressure	m	36.6
3	Engine power	kW	4

Diesel fuel is heated by electric heater mounted in tank with diesel fuel.

Diesel fuel is pumped into combustors of gas turbines from the tank through fuel filters.

Technical specification of transfer pump of diesel fuel is described in the Table 10.3.17.

Table 10.3.17 - Technical specification of transfer pump of diesel fuel

Ser. No.	Name	Size	Value
1	Production rate	t/h	37.2
2	Pressure	m	90.1
3	Engine power	kW	18.5

10.3.3.12. PROCESS AND INSTRUMENT AIR SYSTEM

Plant air is produced in compressor of integrated gas chemical complex and is fed to a gas turbine power station through pipelines.

Plant air is fed to the compressed air receiver. Compressed air goes from receiver to consumers.

Technical specification of compressed air receiver is described in the Table 10.3.18.

Table 10.3.18 - Technical specification of compressed air receiver

Ser. No.	Name	Size	Value
1	Volume	m ³	30
2	Pressure	bar	8

Plant air is divided into instrument and process air.

Process design of instrument and process air provides following flows:

- instrument air feed from IPC to air storage receiver
- instrument air feed from receiver to gas turbines 1 - 4
- instrument air feed from receiver to steam turbines 1, 2
- instrument air feed to condensate

- instrument air feed to service building
- instrument air feed to exhaust heat boilers 1 - 4
- instrument air feed to condensate treatment
- instrument air feed to deaerator
- process air feed from IPC
- process air feed to gas turbines 1 - 4
- process air feed to steam turbines 1, 2
- process air feed to condensate
- process air feed to exhaust heat boilers 1 - 4
- process air feed to condensate treatment
- process air feed to deaerator
- process air feed to service building.

10.3.3.13. PIPELINES, VALVES

Pipelines and valves used in the project are made of steel:

- high-pressure pipelines and valves – alloyed steel;
- low-pressure pipelines and valves – carbon steel;
- pipelines of treated water – carbon steel with an internal anti-corrosion coating

10.3.3.14 THERMAL INSULATION

Equipment and pipelines inside building with temperature above 45 ° C and pipelines outside the building with temperature above 60 ° C should be insulated thermally.

Equipment and pipelines with environment temperature below 60 ° C outside the building should be insulated thermally to prevent freezing.

Cooling water pipelines inside the building should be thermally insulated to prevent condensation on their surface.

The surface temperature of insulation for pipelines and equipment inside the building with temperature above 100 ° C, does not exceed + 45 ° C.

The surface temperature of insulation for outside pipelines (with metallic coating layer) does not exceed + 55 ° C.

Items made of mineral-cotton extra thin basalt fiber shall be used as a main heat-insulating layer. Galvanized steel is used as a topcoat.

Insulation is performed according to mounting technology of an organization.

10.3.3.15. LAYOUT ARRANGEMENT

Main equipment of gas turbine power station (gas turbines, exhaust heat boilers, steam turbine, deaerator, feed water pumps of exhaust heat boilers, feed water pumps for production needs, demineralized water pumps, diesel fuel pumps, intermittent and continuous blowdown separators of exhaust heat boilers, compressed air receiver) is located in a building with area 216x74 m.

The floor height of 0.000 m of the main building is accepted an absolute mark of 100 m.

Deaerator is located between exhaust heat boilers No.1 and No.2 at 14.000 m.

Metal exhaust stacks of exhaust heat boilers with 2,900 mm diameter and 70 m height are located outside the main building on A row side. Bypass metal exhaust stacks with 2,900 mm diameter and 70 m height are located inside the main building.

Air condensers are located in an open area with size 38.7 x 42.045 m.

Condensate pumps of steam turbine are located under the air condensates at 0.000 m level.

Tanks of diesel fuel and demineralized water are located in an open area on A row side of the main building. Gas inlet unit is located in the end of a main body.

10.3.3.16. REPAIR OPERATION MECHANIZATION

There is a travelling crane with lifting capacity of 100/20 tf, span - 25 m, 14 m - lifting height for mechanization of repairs in an engine room.

10.3.4. WATER-CHEMISTRY CONDITIONS

10.3.4.1. DESALTING PLANT

Demineralized (desalted) water is produced by a desalting unit located in a separate building of water treatment for general use of the Integrated Petrochemical Complex (IPC).

Drinking water from households is a source (raw) water going to desalting plant.

Demineralized (desalted) water is treated through following steps: Filtering by mechanical filters, by H-cation filters, water degassing by calciner, filtering by anion filters followed by final desalting on mixed bed filters (MBF).

Deep desalted (demineralized) water is supplied through two pipelines to a storage tank with demineralized water.

Desalted plant and water chemistry conditions provide standard indicators of quality feed water of boilers in accordance with the paragraph 4.8.23 of the Rules on technical operation of power plants and networks of the Republic of Kazakhstan, approved by the Government of the RK on 24.10.12, No. 1352.

10.3.4.2 STORAGE AND DISTRIBUTION OF DEMINERALIZED WATER

Following equipment and piping designed for storage and distribution of demineralized water:

- storage tank for demineralized water with diameter of 15,300 mm, height - 11,400 mm, and operation volume - 1753 m³;
- two demineralized water pumps (one working, one backup);
- recirculation lines for pumps, including control valve;
- piping, valves and instrumentation for distribution of demineralized water to all consumers within plant.

Demineralized water is stored in a storage reservoir with atmospheric pressure. There is provided a nitric protection to prevent contact of water with oxygen environment.

The reservoir is enveloped with nitrogen and equipped with a vent line exiting into the atmosphere; drains of tank and drain lines flow to the floor drain system. The tank is equipped with an indicator of a local level, level sensor, and temperature sensor.

Demineralized water pumps are connected to the tank with collectors, which have a multi-way valve near the reservoir.

Collector is connected with suction pipes of demineralized water pumps, each of them has a multi-way valve, deep cleaning filter, and device to measure local pressure.

The system is equipped with a recycle line with a minimum consumption (from discharge pump to reservoir tank) with control valve, which receives a signal from a flow sensor located in general discharge manifold pump.

Lines go to consumers from pressure collector.

Tank is equipped with level gages preventing emptying tank with demineralized water and running the pumps if the pressure level is low.

The feeding line of demineralized water is equipped with control valve for regulation water level in the tank.

Demineralized water supply system from existing desalting unit is performed by automatic controlling level or manually by operator in control room.

10.3.4.3 SYSTEM OF CHEMICALS INTRODUCTION INTO WATER-STEAM CYCLE

System of chemicals introduction maintains standard indicators of steam and water quality.

System of chemicals introduction consists of three main metering systems:

- introduction of ammonia from PNL B.L. to the intake manifold of two feed water pumps of heat recovery steam generator and boiler feed water pumps to the PNL to achieve required pH;
- oxygen scavenger (hydrazine hydrate) dosing unit to intake manifold of two feed water pumps of heat recovery steam generator and boiler feed water pumps to the PNL
- trisodium phosphate dosing unit to adjust pH at two steam receivers for heat recovery steam generator.

Introduction of ammonia is used to maintain the pH level of the condensate/feed water by inputting ammonia solution.

Ammonia solution goes from PNL through pipelines.

Ammonia introduction system includes internal piping with all valves, accessories and instrumentation.

Flow rate of ammonia solution is 2-3 kg/hour. Ammonia flow amount can be automatically or manually adjusted by feed pump control in the DNL.

An introduction unit of oxygen scavenger is used to maintain required oxygen content in condensate/feed water by inputting solution through oxygen absorber (hydrazine hydrate).

Oxygen absorber unit includes following equipment:

- two controlled volume pumps (one working, one backup) with electric engines (solution of hydrazine hydrate for feed water pump inlet of steam generators and heat exchangers);
- two controlled volume pumps (one working, one backup) with electric engines (solution of hydrazine hydrate for feed water suction pumps of DNL boiler);
- tank with 1 m³ capacity, made out of stainless steel, equipped with gauge, switch and others;
- discharge pump with air drive;
- two calibration tank (one per line) with relevant shutoff valves;
- internal piping with all valves, required accessories and instrumentation.

Before starting oxygen scavenger unit the container should be filled by operator manually by discharge pump with pneumatic drive.

The transmitter inside the container indicates the level to the operator during loading hydrazine hydrate solution.

Only one controlled volume pump for each line operates in normal condition, while the other is used for backup. If working pump is out of service, the backup pump starts automatically (by switching).

Hydrazine hydrate solution is introduced at a rate of 0.12 l/h in normal continuous operation of feed water pumps of steam generator and heat exchangers, and 0.31 l/h in normal continuous operation of boiler feed water pumps.

Hydrazine hydrate solution feed can be performed automatically or manually by adjusting the stroke of controlled volume pumps proportionally to water flow through feed pumps.

Phosphating of boiler water prevents boiler from calcium scale.

Trisodium input unit is used to maintain required steam pH by introducing trisodium phosphate solution.

Trisodium phosphate input unit includes following equipment:

- three controlled volume pumps (two operating, one backup) with electric engine for boiler drum of steam generators and heat exchangers;
- tank completed with gauge, switch and other;
- discharge pump with pneumatic drive;
- two calibration tanks (one per line) with respective shutoff valves;
- internal piping with all valves, accessories and instrumentation.

Reagent goes to phosphate unit as a solution in barrels up to 200 l.

The operator using discharge pump with pneumatic drive manually before starting the trisodium phosphate unit should fill out tank.

The transmitter inside the container indicates the level to an operator during loading solution of trisodium phosphate.

Only two controlled volume pumps operate in normal condition, while the other is used for backup. If one of working pump is out of service, the backup pump starts automatically (by switching).

Trisodium phosphate solution is introduced at a rate of 0.18 l/h in normal continuous operation of each boiler drum (one for each steam generator and heat exchanger).

SECTION 10.3.5. AUTOMATED CONTROL SYSTEMS (ACS)

10.3.5.1. GENERAL INFORMATION

Automation section is developed based on decisions taken in engineering part of “Construction of infrastructure facilities of special economic zone “National industrial and petrochemical techno park” project.

Gas turbine power station consists of four gas turbine generators (GTG), four exhaust heat boilers (HRSG), and two steam turbine generators (STG).

Gas turbine power station is designed to optimize efficiency in base-load operation, with sufficient modularity and flexibility in different conditions, or at partly load.

Construction power station allows resistance to internal and external faults for:

- maintaining output power, if applicable;
- fault repairing minimizing loss of output power;
- maintaining power plant in condition, which allows minimizing time required to resume production after troubleshooting.

10.3.5.2. AUTOMATED CONTROL SYSTEMS OF OPERATING PROCEDURES

Automated control system of operation procedures (ACS OP) is designed for automated control and management plant operation procedures, preparing and storing database in real time, and for data exchange via corporate information computer network (CCN) with an automated control system of upper level. ACS OP is created based on modern digital control system using microprocessor technology with self-diagnostics hardware, redundant controllers and input-output units.

Developed ACS OP should comply with general requirements established by all-Union State Standard 24.104-85 “Single standards of automated control systems. “Automated control systems. General requirements”, and with requirements set out in following documents:

- Standards of the RK 34.014 -2002 “Information technology. Set of standards of automated systems. Automated systems. Terms and definitions”.
- all-Union State Standard 24.701-86 “Single standards of automated control systems” “Reliability of automated control systems. General provisions”.

- IEC 61508 “Functional security of electrical, electronic, programming electronic networks related to security issues”.

- “Rules of technical operation of power plants and networks of the Republic of Kazakhstan” approved by Government of the RK on 24.10.12, No. 1352.

- RD153-35.127-2002 “General technical requirements to software and hardware complex for ACS OP thermal power plants”

- RD 153-34.1-35.104-2001 “Methodical instructions on volume of process measurements, alarms, automatic control in thermal power plants CCGT unit equipped with ACS OP”

- RD 153-34.1-35.137-2000 RD 153-34.1-35.137-2000 “Technical requirements to the subsystem technological protections based on microprocessor technology”

ACS OP is a system of distributed information management human-machine, designed for a long-term operation in a real time, which includes operational and technological operational personnel, software and hardware complex.

ACS OP has high resiliency and provides reliable, efficient and safe operation of facility that meets design parameters on products performance and quality, consumption of fuel and energy resources, extending overhaul life and improving environmental safety.

ACS OP is designed to perform a set of information and control functions to provide following:

- control with an appropriate representation of information at operator-technologist workplace with parameters necessary for normal and safe conduction of technological process;
- ongoing analysis of parameter changes to critical values and prediction potential accidents;
- automatic control over process parameters;
- pre-alarm and warning process failures;
- registration of operation and control over devices of emergency automatic protection;
- protection of process and electrical equipment during system failure;
- automatic diagnostics of equipment;
- processing accounting parameters, calculation of material balance and technical and economic performance of facility;
- storing process data for a certain period and creating data files;
- printing out necessary information about technical process and its safety;
- monitoring air quality within facility;
- providing information about facility protection to the superior system.

10.3.5.3. Composition of ACS OP

ACS OP includes following:

- Distributed Control System (DCS);
- Emergency Shutdown System (ESD);
- Fire and Gas detection and protection system (FGS);
- Burner Management System (BMS);

DCS, ESD, FGS, BMS should be independent by hardware, software, and power supply from other automation systems.

Other secondary diagnostic systems and equipment maintenance:

- Continuous Emission Monitoring System (CEMS),
- Machine Management System (MMS),
- Gas Turbine Control System (GTCS),

- Steam Turbine Control System (STCS),
- Power Plant Performance Management System (PPMS),
- Equipment management operations system, instruments and controls (IAMS)
- Package (local) control systems etc.

All systems are future with expansion ability by availability of reserve hardware and space for extra modules.

ACS OP includes following levels:

- first – filed instruments and controls based on modern electronic technology and local controller (converters and interfaces)
- second – programmable logic controller (PLC) of distributed control systems and automated workstation of operator-technologist (OWS).

Backup buses of data exchange perform information exchange between PLC technology subsystems and OWS automatically. Data exchange buses are interchangeable with determined protocol and transmission rate not less than 4 MB/s.

Communication between first and second levels components of management is carried by codes, analog and digital electrical signals.

Specialized industrial computer networks of high performance, providing full cycle of data exchange spending no more than 3 seconds carry out data exchange between first and second levels components of management.

ACS OP subsystems have hardware and software diagnostics of information networks, OWS, blocks and modules, as well as input and output circuits. Status information is transmitted to the OWS for display and recording.

Information is protected from unauthorized access to ensure normal operation of ACS OP and prevent failures during operation processes. Protection is provided by software password so that:

- every user can access the ACS OP by using password only
- there are various levels of access for different types of users
- every user has certain type of capability set to observe or change data according to his/her access level.

ACS OP software (SW) is protected from unauthorized access by copyright for software products, software hardware to control integrity of SW and software hardware to identify users. Moreover, software OWS includes program protection against viruses.

ACS OP information integrity is provided by using special storage technology, which ensures preservation and recovery of retrospective information.

Information about emergency is automatically displayed at OWS, recorded and stored in records to message system on devices with non-volatile memory. If communication devices fail, the second level components of ACS OP will process independently. Failure message will appear on OWS. After fault recovery of communication devices the exchange information is restored automatically with an appropriate message on ACS OP.

In addition to OWS, the system has engineering workstation (EWS), which performs engineering functions in ACS OP operation.

Electric power station is controlled by distributed control system (DCS) during basic operation, which uses all logic and consistency. In case of accident, emergency and safety interlocks implemented in the system of emergency shutdown (ESD) take control.

10.3.5.3.1 DISTRIBUTED CONTROL SYSTEM (DCS).

Distributed control system has modular architecture with maximum modular interchangeability. To connect modules the high-speed communication network with reservation has been accepted.

The system configuration is designed so:

- failure of one module does not affect the functional capabilities of other modules;
- failure in data communication channel does not affect the operational capability of microprocessor controllers;
- failure of control unit at one operator does not affect the operational capability of other system control units nor limits control and monitoring system functions;
- each control unit together with electronics is autonomous;
- failure of one datastorage module does not result in loss of registration functions or industrial data;
- failure of power supply module does not result in control failure;

Controllers with reservation carry out control and related functions. The system continuously controls the readiness of reserved controllers and any failure has its own indication. The main and reserved controllers are identical. In case of failure of working and reserved controllers, the system runs down in the last position or transfers to the specified conditions.

DCS controllers provide self-check with determination of internal module failure and failure in hookup of input/output circuit of PCI with indication and failure registration in OWS.

It is possible to remove and replace any input-output board without supply disconnection and process interruption by using reserved manned station.

Analog DCS modules carry data transfer via HART-protocol.

DCS consists of historical server and archiving unit as well as active spark protection barriers.

DCS provides at least the following functions:

- automatic control;
- indication of pilot circuit condition, including input process information, controlled value, output parameter, and controller operating mode in the form of bar chart and in digital form;
- calculation of process control parameters;
- switching control circuit to the manual operation as well as changing controlled values, modes and control parameters from OWS;
- indication of changeable processes;
- registration of changeable process trends (with switching capacity), including controlled and estimated variables;
- control of isolation valve, rotating equipment, and status indication
- indication and audio announcement about emergency situation as well as system failure;
- data collection by use of observed values and calculations in data base;
- reporting on request based on current and archival data;
- blocking and blocking priority management to prevent emergency situations;
- registration of alarm notifications and system events;
- colored graphic display of installation area, including control loops and process information in real time;
- system diagnostics.

The system provides operator interface with reservation and usage of independent electronic equipment, OWS and system interface facilities.

Operator interface allows:

- visualize process parameters on-line by videogram, graphics and trends;

- process received information;
- carry out diagnostic tests, change set values of control constants from operator control panel;
- transfer from automatic mode to manual control of control and cutoff devices;
- switch off electric equipment remotely;
- print document copies, protocols, logbook, etc.

10.3.5.3.2 EMERGENCY SHUTDOWN SYSTEM (ESD)

ESD is of the high-priority towards simple control facilities and allows reducing hazard for personnel and volume of equipment damage, process violations and environmental damage.

ESD activated manually or automatically provides remote equipment shutdown, isolation and prevention from escalation. The system switches off process equipment, which can contribute to hazardous behavior.

ESD is actualized on the base of programmable logic controllers aimed to continuous control of extremely significant operational parameters, programme control of isolation valve and equipment security.

ESD provides:

- secure running of operational process;
- recording of source of blocking and storing the sequence of actuation mechanisms and personnel actions in case of emergency situations;
- management of electric equipment and automated isolation valve to prevent escalation of emergency situations;
- self-check of system units with status indication.

ESD is performed as fail-safe system with reserved central process. All analog inputs, digital inputs and outputs, data bus lines and power supply sources are reserved.

In case of process emergency, supply disconnection or loss of compressed air supply of CI, ESD provides transfer of production facility to safety condition. Return to operation condition after ESD actuation is carried out by operating personnel according to instruction.

System and applied software of ESD uses only standard program methods such as functional logic schemes.

Adjustment of alarm level for analog inputs, timers, operating mode of contact input/output are carried out programmatically.

There is an unauthorized access protection in ESD by settling password hierarchy for users.

ESD operates independently from other systems. ESD is integrated with DCS and unitized by reserved bar. These communication lines are reserved and provide integrality of system connection so neither failure affects the other one. Readiness of communication line is checked permanently and in case of any failure, the warning is displayed in OWS.

Input/output lines in serial communication between ESD and DCS are organized on the following principles:

- for each analog and contact input- transfer of emergency signal to DCS;
- for each contact output- signaling on output status to DCS;
- for each programme input of interlock bypass- addressing to DCS about status of interlock bypass;
- for each switching of interlock bypass- addressing to DCS about status of switching;
- for each contact input operational interlock bypass- addressing to DCS about its status;
- each remote input/on/off input from DCS shall be displayed in ESD;
- emergency power supply signal and system emergency signal of ESD shall be submitted to DCS;

ESD has alarm formation facilities. Part of alarm goes to actuation safety mechanisms.

The other part of alarm DCS gradually processes and give relevant instructions on process equipment management.

Registration of ESD signals is carried out by each operator station (Historical Report).

As ESD shall operate in single time scale with DCS, these systems have time synchronization unit.

10.3.5.3.3. Fire and gas detection and protection system (FGS).

FGS includes gas pollution detection system, fire and fume detection system, fire-control system, relevant alerting system, etc.

FGS shall be separated from DCS and ESD under hardware control. The connection shall be provided by reserved bar.

10.3.5.3.4. BURNER MANAGEMENT SYSTEM (BMS).

BMS controls and manages burning process in exhaust heat boiler (HRSG) and includes fuel and air supply system, fire control system, CI air supply system to the actuation mechanism, etc.

All field sensors and actuation mechanisms involved in the control process of burning management are connected to the separate BMS controllers.

BMS shall be separated from DCS and ESD under hardware control. Connection shall be provided by reserved bar.

10.3.5.3.5. CONTINUOUS GAS EMISSION MANAGEMENT SYSTEM (CEMS).

The system is primary designed to monitor environment pollution by nitrogen oxide emission (NO_x), carbon oxide (CO) and dust from burnt gas section.

As the base solution two points of sampling is provided for each CEMS: the first- to the main funnel chimney, the second- in by-pass one, in particular:

- CEMS No. 1 for exhaust heat boiler No.1 (HRSG No.1) – main funnel chimney and bypass funnel chimney;
- CEMS No. 2 for exhaust heat boiler No.2 (HRSG No.2) – main funnel chimney and bypass funnel chimney.
- CEMS No. 3 for exhaust heat boiler No.3 (HRSG No.3) – main funnel chimney and bypass funnel chimney.
- CEMS No. 4 for exhaust heat boiler No.4 (HRSG No.4) – main funnel chimney and bypass funnel chimney.

The system in both funnel chimneys (main and bypass) shall continuously measure the following burnt gas parameters:

- CO - NO_x - O₂;
- dust concentration;
- consumption;
- absolute pressure;
- temperature.

All components and materials required for analysis (excluding sampling filter, heated sampling pipeline, equipment for dust level measurement and consumption/pressure/temperature measurement devices) shall be installed, connected by pipes and cables inside the analyzer box (shelter) which is applicable for installation in secure area.

In sampling place installed in funnel chimney there shall be provided the place for installation of temperature, consumption, dust emission, and pressure sensors.

Analysis system consists of the following parts:

- analyzer box (shelter);
- gas sampling system: sampling device with filter and heated sampling pipe line;
- gas sample conditioning system;
- analyzer module for flue gas analysis (CO, NO_x, O₂);
- equipment in funnel chimneys for flue gas analysis (dust concentration, consumption, pressure, and temperature);
- data collection and process system;
- standardized gas bottle and pressure regulator.

10.3.5.3.6. MACHINERY MANAGEMENT SYSTEM (MMS)

To control machinery technical condition the automated safety resource- saving equipment running management system is applied- machinery management system (MMS), for example, Bently Nevada 3500 or equivalent. Stationary equipment management system is designed to provide safety eco-friendly resource-saving running of process equipment, failure prevention, operating bug, and management of equipment technical state by continuous computer monitoring.

Continuous monitoring excludes pass of unfavorable trend of changing the technical condition of equipment without its suspension and allows to timely detect faults and failures of major units, take prompt measures to recover working capability, cease extension of coming failures.

MMS includes:

- a) distributed system of sensors controlling main parameters of equipment (vibrational acceleration, vibration velocity, vibratory displacement, axial deflection, current, temperature, etc.);
- b) distributed system of remote module providing primary sensor signals conversion and their transfer to the diagnostic controller as well as providing control of integrality of sensors and communication lines;
- c) diagnostic station providing data collection, storage, processing, display of monitoring results;
- d) diagnostic enterprise network to provide at the users' computers full and timely information about technical condition of equipment.

MMS will be a separate system with modular structure and reserved power supply resources.

10.3.5.3.7. INSTRUMENTATION AND AUTOMATED MANAGEMENT SYSTEM (IAMS)

IAMS provides continuous control of status, events and conditions of microprocessor sensors, valve position management devices and other microprocessor devices to create effective system of precautionary maintenance in real time.

IAMS includes elements and characteristics described below:

- Reliability of control devices and valves:

Field devices (sensors and valves) have diagnostic capability with the purpose of precautionary maintenance.

DCS supplier shall take into consideration the following:

- It is necessary to analyze and define for each IC and actuation mechanism (valve) diagnostic methods and opportunities necessary to carry out precautionary maintenance.

The system uses HART protocol for exchange data with peripheral devices for storage of separate unit information, including (but not limited): tag number, description, device type number, serial number, manufacture and warranty information.

The system automatically checks changes of information about field devices and automatically upgrades, where applicable.

IAMS provides continuous control of field devices and sends relevant emergency alarm through DCS when detects any failure. This system includes all diagnostic facilities determining all failures and breakdowns of system hardware to maintain operability of system.

The system is the only facility for remote calibration of intelligent field devices. It consists of relevant software calibration and calibration data of installed intelligent devices (with password protection).

IAMS configuration ensures ESD devices that their reliability will not reduce when interaction with it.

10.3.5.3.8. POWER PLANT MANAGEMENT SYSTEM (PPMS)

ESD shall have the opportunity to control and improve the equipment condition. Online improvement provides various situations and scenario affecting operability, thereby the productivity of power plant.

Online calculation as well as recording of balance parameters, operating and economic indicators of power plant include:

- gas turbine efficiency (GTS);
- exhaust heat boiler efficiency (HRSG);
- steam turbine generator efficiency (STG);
- state estimation and heating limits of exhaust heat boilers (HRSG);
- state estimation and heating limits of steam turbine generators (STG);
- transformer losses;
- plant heat power, gross load and net;
- current losses in comparison with design values.

10.3.5.3.9. TURNKEY (LOCAL) MANAGEMENT SYSTEMS

Turnkey local management systems are applied for supplementary processes in power plant, where decentralized and independent automatic control applied for measurement and control that indirectly involved in normal and continuous work of power plant and where failures do not directly affect the general process of power plant.

Management and control algorithms will be displayed and controlled by PCS, in particular:

- by hardware connection, which includes basic commands (e.g. start-up, stop, etc.) and status signals (e.g. system operation, system failure, etc.). All emergency signals of exchange between management local systems and ESD shall be hardware generated;
- by series connection (reserved only with “important” management local systems which directly affect the general process of power plant) for remote control.

Controllers delivered in sets shall have diagnostic function to facilitate the detection and localize failures.

Failures occurred out of block equipment and which require shut down of equipment shall be directed from PCS to the turnkey local management systems which carry out the relevant shut down..

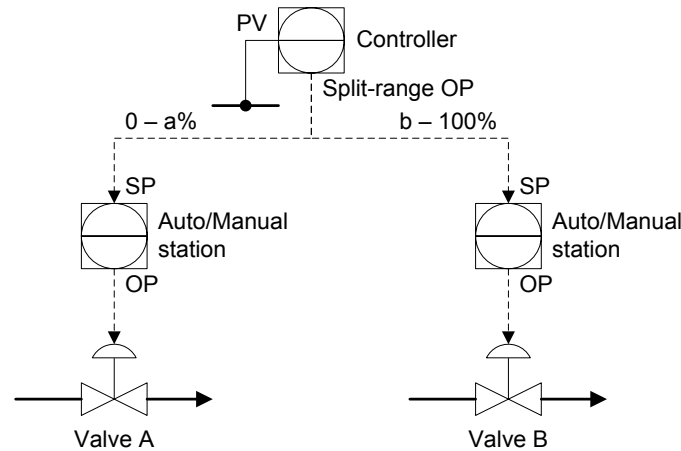
10.3.5.4. TYPICAL CONTROL CIRCUITS

Range division

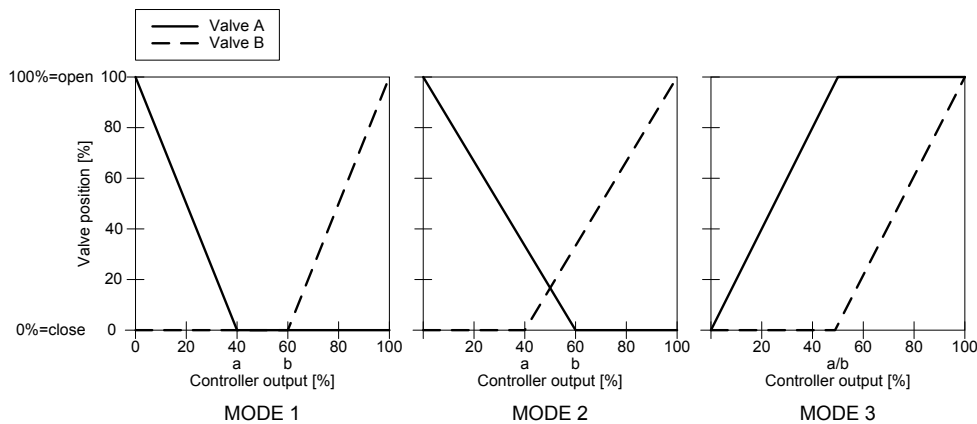
Controller sends 0-100% output to two functional blocks, block 1 and block 2, which operate in particular part of range of controller output. Functional Block 1 converts 0 - a% from regulator output to the signal for valve management (e.g. 100-0% which closes valve when increasing the output signal of controller). Functional block 2 converts b-100% from regulator output to the signal for control valve B (e.g. 0-100% opens valve when increasing the output signal of controller).

There is an opportunity to control cut point from 0 to 100%, not only into 50%. In addition, it is possible to cover cut points. For example, valve A is opened from 0 to 100%, when controller opens from 0 to 52% and valve B is opened from 0 to 100%, when controller opens from 48 to 100%. There is redefining of cover so both valves are closed when they meet in the middle (e.g. when valve is closed from 100% to 0% and valve B is opened from 0% to 100%).

See also graphical form of range division of controller in picture 10.3.1 and various possible split-range of functions in picture 10.3.2.



Picture 10.3.1 – Circuit with range division chart.



Picture 10.3.2 - Modes of range division

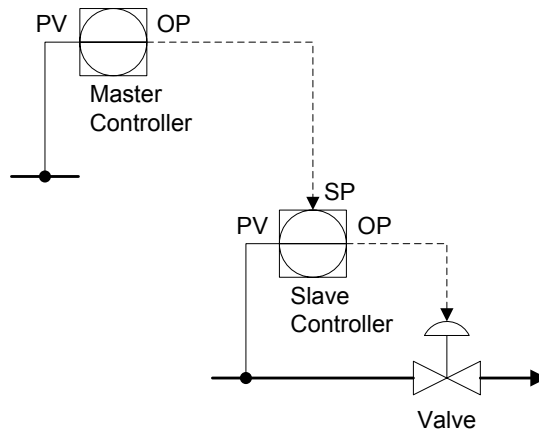
Circuit of cascade control

Mode controller shall be activated one by one in cascade structure. The lowest in the hierarchy controller (usually referred to as slave controller) shall be placed in cascade mode up to the higher controller (usually referred to as master controller).

When slave controller is in CASCADE mode, the master controller in one cycle automatically switches to AUTO mode.

When slave controller has been disengaged from CASCADE mode into AUTO or manual mode, the controller that is higher in the circuit cascade switches to the initialization mode.

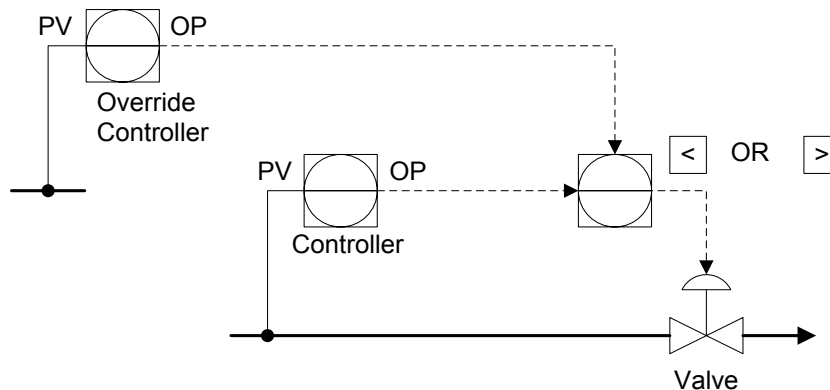
When slave controller in the cascade control circuit reaches its set value, any controller that is higher in the cascade structure shall not continue changing set value of slave controller in the direction that will further exceeding the limit. See also graphic form of master-slave controller in picture 10.3.3.



Picture 10.3.3 – Cascade control circuit chart.

Priority control

Control circuit with redefining of management consists of main controller, one or several management controllers, and priority block. Depending on redefining high and low priority in management, selector selects the biggest or smallest inputs to control valve.



Picture 10.3.4 – Priority control circuit chart

10.3.5.5. AUTOMATION OF HVAC (HVAC)

The project stipulates the usage of turnkey management systems for plenum and exhaust systems. Such management systems provide:

- Automated temperature and humidity maintenance in the premise;
- Status control and remote control of air handling equipment.

There is automatic shutdown of ventilation system in case of fire. The project also provides connection between local controllers of ventilation units and DCS to transfer data about condition of ventilation systems to the operator's control panel.

10.3.5.6. TECHNICAL CONTROL AND AUTOMATED FACILITIES

Control and management of technological parameters is carried out with the use of field automatics and computer equipment.

The project provides control and measurement of:

- temperature;
- pressure;
- pressure ratio;
- level;
- consumption;
- flow analysis of content and quality of flue gas, water steam, steam condensate, boiler feed water, etc.;
- existence of burner flame;
- existence of critical pre-explosion concentration of gas and steam, and limiting concentration of hazardous substances in the air of explosion hazard zones.

Types of received equipment.

To control technical parameters as the basic system the electronic system of devices has been accepted with output current signal 4-20mA with HART protocol. Sensors and actuation mechanisms placed in explosion hazard zones are selected in accordance with group and category of explosives in explosion-proof version with spark-proof circuit (preferably) or explosion-proof enclosure type of explosion protection.

Connection sensors with electric output signal to PCS controllers is carried out through active spark-proof barriers.

Sensors and electronic inverters, as a rule, charged from PCS in two-wire line. Some special control and measuring devices (e.g. analyzers or mass flow meters) which require separate power supply charged from their own power supply source.

All field equipment designed so it can be installed in production equipment and pipelines of outdoor plant under the conditions of climate zone of designed project. Field devices, where applicable, are installed in heated boxes.

Control and measuring equipment placed in open area shall have protection index at least IP65. Local control panels shall have protection index IP55.

Temperature measurement.

- For remote control of temperature mostly resistance temperature device of Pt100 and thermoelectric converter of XK (L) or XA (K) are applied. All preliminary converters are delivered with integrated converters in unified current output 4-20 mA with HART-protocol and flanged protection sleeves.

- For local measurement bimetallic thermometers with flanged protection sleeve is applied.

Measurement of pressure and pressure ration.

- for remote control of pressure converter of excess pressure and pressure ratio of diaphragm type with sensing elements made of stainless steel, in set of two-, three- or five-valve manifold and impulse line binder. To measure pressure of corrosion, viscous and high-temperature environment diaphragm seal or separation vessels with impulse line binder are applied.

- For local measurement industrial pressure gage with rotary connection is applied. To measure pressure of corrosion, viscous and high-temperature environment gages are equipped with diaphragm seal, and, where applicable, cooling agency.

Flow rate measurement.

- For remote control of flow rate mostly flowmeters based on differential measurement method and vortex flowmeters are applied.

- For measurement of electrically conductive environment flow rate electromagnetic flow meters are applied.
- For control of gas, steam, air, nitrogen flow rate differential measurement method with flanged connection to Annubar process is applied.
- For commercial recording mostly Coriolis acceleration flowmeters are applied which provide controlling of density and temperature at the same time. Where conditions do not allow measuring, then one of above mentioned methods is applied with necessary correction in temperature and, if applicable, pressure.
- For local measurement of flow rate LCD indicators integrated in sensors or rate-of-flow meter are applied.

Flowmeters are delivered with counter flanges. The material of flanges will correspond to pipe lines material.

Level measurement.

For remote control of level mostly microwave (reflex-radars) level controllers, radar (horn) level controllers, differential measurement circuit are applied.

Level controllers are installed at bridges, when it is specified by process conditions or design features of device, the installation is allowed in flanged union nipple of capacitor, device.

In-situ measurement is carried out with the use of LCD indicators integrated in sensors and local bypass level indicators.

Analyzers.

To optimal control of technological processes and compliance with ecological norms the power plant is equipped with continuous automated physicochemical analyzers, which allow to receive data with minimum delay, in particular:

Continuous gas emission monitoring system CEMS

- oxygen content in flue gas (O₂);
- carbon oxide (CO) and nitrogen content in flue gas (NO_x);
- dust content in flue gas.

Sampling system

- pH (pH);
- electrical conductivity (COND.);
- cation conductivity (CATION COND.);
- concentration of dissolved oxygen (O₂);
- concentration of silicon oxide (SiO₂);
- sodium concentration (SODIUM);
- phosphates concentration (PHOSPHATE).

Gas contamination control.

Gas contamination control in explosion hazard industrial premises and areas is carried out with the use of gas and steam alarm.

Signals about gas contamination are received in fire and gas detection and protection system (FGS), where it commands to switch on sound and light alarm in gas area and operator room, and signals to switch on emergency ventilation in premises.

Control and shut down valves.

Control and shut down valves with pneumatic drive are used as control and shut down devices correspondingly.

Control and shut down valves are equipped with electropneumatic positioning elements, three-way electromagnetic valves, final position alarm devices and air filters-regulators with the following specifications:

- electropneumatic positioning elements with monitoring signal 4÷20 mA+HART; explosion protection marking “Exi”;
- three-way electromeagnetic valves with monitoring signal 24 V DC, explosion protection marking “Exi” and “Exd”;
- final position alarm devices with explosion protection marking “Exi” or “Exd”;
- filters-regulators are designed for pressure up to 1.0 MPa (10 kgf/cm²).

Valves are delivered with counter flanges. The material of flanges will correspond to pipe lines material.

Automation equipment installation requirements.

Installation of PCI equipment and cables is carried out in accordance with construction norms and regulations 3.05.07-85 “Automation systems”, EIC of RK, approved by the Government of the RK dated October 24, 2012 No.1355 and guide materials of “Montazhavtonatica” JSC.

Connection of sensors and actuation mechanisms is carried out with individual cables from input packs up to intermediate junction boxes. Trunk multi-circuit cabling is laid from intermediate boxes to jumper cabinets.

Measuring spark-proof circuits and non-spark-proof ones are laid in separate boxes. In controlling room under raised floor spark-proof and non-spark proof cables are also laid separately.

To install sensors in industrial pipelines and union nipples integration technological part of project provides base constructions and fundamental isolation valve.

Calibrated pipe is used for impulse lines, for example, Ø14x2, and connectors made of stainless steel.

Pulse tubes with measuring environment which needs heating are heated from sampling place to separation vessels or sensors placed in heated cabinets of CI. Heating of impulse lines is carried out by self-regulating electric cable. Electrical heating of CI cabinets – integrated electric heaters with thermostat.

Power supply of actuation mechanisms of control and shut down valves is provided from CI air receiver.

Distribution of air to consumers is carried out by tubes made of cold-resistant steel. Direct connection of positioning element or solenoid is carried out with the use of calibrated tube, for example, Ø8x1, made of stainless steel.

All automated facilities with voltage are grounded. The following shall be grounded:

- metallic cases of devices and automated facilities, management and alarm devices;
- metallic boards to install PCI and PCS facilities;
- metallic housing and control cable displays, cable structures.

10.3.5.7. CI AIR

Air source for consumers of control and measuring devices shall comply with the following:

Indicated pressure 6.5 bar

Dewpoint temperature: not lower than -63°C under atmospheric pressure.

The air shall not contain dust and oil as well as corrosion and toxic gas and shall be of 1 contamination class under GOST17433-80.

10.3.5.8. INSTRUMENTS AND CONTROLS ELECTRIC SUPPLY

Working power supply goes from AC network with 230 V voltage and 50 Hz frequency.

To provide nonfailure operation of electric station PCS and field instruments and controls there are two duplicate uninterruptible power supplies (UPS) 230 VAC, 50 Hz. Uninterruptible power supplies shall provide power backup during 120 minutes.

Supply of sensors circuit with 24 VAC is made from reserved supply nits which are installed in power distribution cabinets 24 VAC.

Fire and gas detection and protection system (FGS) shall operate from power backup system; time of power backup provision is 24 hours. Power backup battery pack and battery shall be supplied together with fire and gas system.

10.3.5.9. CABLING

For reliable operation of all PCS technical means there shall be a reliable structure of communication channels between PCS technical means.

To connect local controllers with sensors and electric actuators physical links shall be provided – copper-conductor cable with required cross-section and in required version: shielded, enclosed, etc.

To connect local controllers with command consoles digital channels are provided. For this purpose fiber-optic cables as well as cables with shielded twisted pair are used.

Cabling is performed over metal structures, equipment, on bridges, in channels and in ground trenches.

10.3.5.10. SAFETY EARTH

Technical means of PCS are complicated microprocessor units.

High rate of data processing and transfer, relatively low level of signals within the system place rigid requirements to earth system which provides protection of all system devices and elimination of interferences.

Necessity to arrange efficient earth is determined by the following basic requirements:

- provision of staff safety against dangerous impact of current;
- elimination of interference, occurring as a result of pickups and causing malfunction of electronic devices operation.

To provide staff protection in accordance with the requirements of GOST 12.2.007.0-75*, GOST 12.1.019-79*, GOST 12.1.030-81* industrial object with installed automatic control system shall be protected with protective earth edge 3.0 Ohm.

To protect instruments and controls against interferences within the scheme instrument grounding bus is provided.

Instrument earth edge is connected with protective earth edge in one point – in point of installation of earth rod. Instrument earth edge resistance shall not exceed 3.0 Ohm.

Installation of protection earth shall be performed in accordance with the guide on installation of protection grounding and neutral earthing TI 4.25088,17000 GOST 12.1.030-81 “Occupational safety standards system. Electric safety. Grounding. Neutral earthing” and EIC RoK approved by the decree of the Government of RK d/d 24 October 2012 № 1355.

10.3.5.11. SAFETY STANDARDS AND MEASURES FOR EXPLOSION AND FIRE SAFETY

Safety standards are observed in two aspects:

safety standards during installation, commissioning and start-up;

safety standards during operation of the equipment.

During installation, commissioning and start-up it is necessary to deenergize the equipment which has “Do not turn on. People at work” signs on their circuit breakers

All equipment which is installed in explosion-hazard area has explosion-proof design or spark-safe circuits.

For equipment which carries voltage during operation grounding edge is provided.

All rooms where people work are equipped with blowing, exhaust and emergency boost extract ventilation. Disabling of working and stand-by fans is performed. Near entrances where occurrence of gas is possible there are signaling devices and buttons to turn on emergency fans at place. Signals on condition of all ventilation systems are shown on corresponding PCS consoles.

10.3.6. ELECTROTECHNICAL SOLUTIONS.

10.3.6.1. GENERAL INFORMATION

Electric part of the project for gas turbine power plant for “National industrial petrochemical technological park in Atyrau region” is developed on the basis of:

- technical specifications
- design engineering assignment
- general plan
- thermal and mechanical and architectural and building sections of the project.

The project was developed in accordance with the requirements of regulation documents.

10.3.6.2. POWER PLANT TO CIRCUIT CONNECTION UNIT

Connection of gas turbine power plant to the system and power output is performed via two aerial circuits AL 110 kV.

Connection to power system of the RK is provided in accordance to TS № 24-02-15/6083 d/d 25.09.2013, issued by «KEGOC» JSC.

AL-110 kV are made of double-circuit line using AC wire. The type of supports, isolation and line hardware of AL 110 kV as well as wire cross section will be determined by working design.

AL-110 kV shall be equipped with protective earth wire with built-in FOL for efficient data transfer via relay and emergency shutdown device of energy-consuming and energy-transmitting sources.

10.3.6.3 MAIN ELECTRIC CIRCUIT

10.3.6.3.1 CIRCUIT DESCRIPTION

Main electric circuit of power distribution for object of social economic zone «National industrial petrochemical technological park» in Atyrau region you can see in the drawing № D019-14-TЭО-ЭМ page 1.

For basic power consumers at the site the project provides electric supply with reliability rating 1. For essential service and for consumers securing safety additional electric supply from independent sources is provided.

Overall load of SEZ NIPTP equals to 248.3 MW. Power plant capacity is $S_H = 310$ MW. Coefficient of network performance is $\cos \phi = 0.9$.

Electric power is produced by electric generators ($U_H = 10.5$ kV) regulated by four gas turbines and two steam turbines. Power plant generators are connected to unit transformers 10/35 kV by complete isolated bus ducts (CIBD).

Table 10.3.5 – Technical specifications of gas turbine generator (4 pcs.).

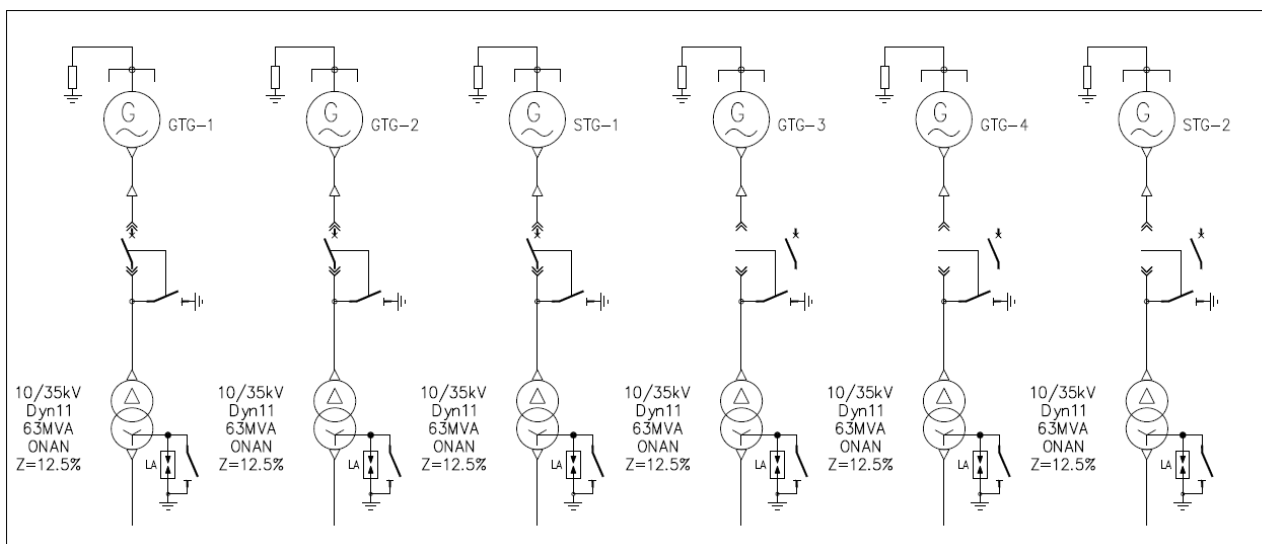
No.	Name	UM	Value
1	Electric power	MW	50
2	Voltage	kV	10.5
3	Number of phases	-	3
4	Number of poles	-	2

5	Rotation speed	rpm	3000
6	Current frequency	Hz	50
7	Total weight	tons	70.3
8	Rotor weight	tons	20

Table 10.3.6 - Technical specifications of steam turbine generator (2 pcs.).

No.	Name	UM	Value
1	Electric power	MW	55
2	Voltage	kV	10.5
3	Number of phases	-	3
4	Number of poles	-	2
5	Rotation speed	rpm	3000
6	Frequency	Hz	50
7	Total weight	tons	87
8	Rotor weight	tons	15

Power supply of the complex is performed through unit transformers 10/35 kV 63 MVA, which are connected to closed type main switcher 3ГРУ-35 kV.



Connection with electric system of the RK is performed with the help of two lines 110 kV and a step-down transformer 110/10 kV.

Power supply circuit of SEZ NIPTP with two systems of collecting bars of main switcher ГРУ-35 kV of the power plant can provide consumers with energy of electric reliability rating 1.

Auxiliary emergency diesel generator is provided at the power plant for energy supply in case of emergency.

Power plant load is equally distributed between the buses at the main switch 3ГРУ 35 kV and medium voltage switchgears 3РУ 10 kV. Main sources, main transformers, switchers and isolating switches in every section of the power plant are designed in view of provision with energy during actual and further possible loads with 20% reserve of power for possible future increased needs.

Switch step-down transformer plants are located in various places of load buildup and shall supply energy via low voltage cables laid on cable structures. High and medium voltage cables are laid below the ground.

Transformer plants are located as close as possible to the places of possible load. Switchers of medium and low voltage plants supply motors 10 kV, inputs 0.69 kV and 0.4 kV. Bus-tie switches are controlled by automatic transfer switch (ATS). Every disconnect switch is provided for full load in case of malfunction of the second switch in other section. Thus, transformers are designed for full load. Switches and transformers have 20% excess power for further development.

Voltage level

The following definition will be applicable:

- LV (low voltage) : up to and including 1 kV A.C. and 1.5 kV D.C.
- MV (medium voltage) : up to and including 72.5kV A.C.
- HV (high voltage) : over 72.5 kV A.C.

Original voltage level : 110 kV, 3Ø, 50Hz, 3 wires, grounding according to standards of RoK.

- Distribution, MV voltage level: 10 kV, 3Ø, 50Hz, 3 wires, grounded resistance
- Distribution, LV voltage level: 690V, 3Ø, 50Hz, 3 wires, direct grounding
- Distribution, LV voltage level: 400/230V, 3Ø, 50Hz, 4 wires, direct grounding.

Standard rated distribution of voltage level is applied as follows:

- 110 kV $\pm$ 5%, 3 phases, 3 wires, 50 Hz $\pm$ 2%, solidly-grounded;
- 35 kV $\pm$ 5%, 3 phases, 3 wires, 50 Hz $\pm$ 2%, non-grounded (only to site);
- 35 kV $\pm$ 5%, 3 phases, 3 wires, 50 Hz $\pm$ 2%, grounding through resistance;
- 10 kV $\pm$ 10%, 3 phases, 3 wires, 50 Hz $\pm$ 2%, grounding through resistance;
- 6 kV $\pm$ 5%, 3 phases, 3 wires, 50 Hz $\pm$ 2%, grounding through resistance;
- 660 V $\pm$ 5%, 3 phases, 3 wires, 50 Hz $\pm$ 2%, solidly-grounded;
- 380/220 V $\pm$ 5%, 3 phases, 4 wires, 50 Hz $\pm$ 2%, solidly-grounded;
- 220 V $\pm$ 5%, 1 phases, 2 wires, 50 Hz $\pm$ 2%, solidly-grounded(I), 220 V $\pm$ 1%, 1 phase, 2 wires, 50 Hz $\pm$ 2% from UPS system, solidly-grounded;
- 220V direct current $\pm$ 10%, 2 wires from rectifying device / battery;
- 125 V direct current $\pm$ 10%, 2 wires from rectifying device system / battery, non-grounded (only additional/emergency supply of the unit within turbine set);
- 110 V $\pm$ 5%, 1 phase, 2 wires, 50 Hz $\pm$ 2%, solidly-grounded(only additional/emergency supply)

Application of motors:

- Motors above 400 kW : 10 kV, 3Ø, 50Hz,
- Motors from 50 kW to 400 kW : 690 V, 3Ø, 50Hz,
- Motors from 0 kW to 50 kW : 400 V, 3Ø, 50Hz,

Other devices:

- Lighting : 230Vac, 1Ø, 50Hz
- Inconsiderable interruptions of circuit and switch : 230Vac, 1Ø, 50Hz
- Heat trace circuit : 230Vac, 1Ø, 50Hz
- Control circuits – MV motors : 110Vdc
- Control circuits – LV motors : 230Vac, 1Ø, 50Hz

- Control circuits – Instrumentalization : 230Vac, 1Ø, 50Hz
- UPS instrument power : 230Vac, 1Ø, 50Hz, non-grounded
- DC power – electric : 110Vdc, non-grounded
- DC power– instrumental : 24Vdc, non-grounded

Acceptable tolerance

- Voltage change in normal conditions : $\pm 5\%$
- Frequency change in normal conditions : $\pm 1\%$

Voltage change

- Energy feeder (plant to plant, distributor to distributor) : 1% of impressed voltage
- Subfeeder : 5% of impressed voltage
- Electric motor – work : 5% of rated motor voltage
- Electric motor – start : 15% of motor voltage
- Electric motor – lighting : 5% of voltage in lamp

Coefficient of performance (COS ϕ) of the whole system including idle power waste in transformers and other equipment of distribution system is not less than 0.90 over the whole electric power plant. If necessary, correction of COS can be achieved by regulating generator reactive capability.

10.3.6.3.2 SHORT CIRCUIT CURRENTS

Incoming power source OHL 110 kV.

Based on client's preliminary data maximum values of short circuit are:

$S''_{kmax.} = 7621.02 \text{ kVA}$ (corresponds to 40 kA,)

Ratio of reactive resistance to direct current resistance is $X/R=10$)

Results of short circuit examination are within the limits of selected equipment.

10.3.6.4 AUXILIARY WIRING DIAGRAM

10.3.6.4.1. DIAGRAM DESCRIPTION

Alternating current supply of auxiliary electric motors:

10 kV for electric motors 530 kW – 710 kW;

690 V for electric motors– 400 kW;

380 V for electric motors less than– 400 kW.

Direct current supplies emergency lighting network, operative regulation circuit, automatic equipment, protection and alarm of main circuit elements and PCS devices.

For powerhouse auxiliary drive induction motors with cage rotor are applied.

To regulate electric motors rotation frequency changers are applied.

Variable speed drive of low voltage electric motors (VSD HH) is provided to regulate operation of cooler / axial-flow fan and to control some pumps. It is recommended not to install them at motor control center (MCC), if they are not small single devices but to assemble them into the groups in panels of «cabinet» type which have common bus and energize from the switch of HH circuit breaker panel.

Panels are distributed over the whole distribution bus of HH circuit breaker panel providing cooling in all supply conditions. Certain variable speed drives of electric motors (VSD) do not require forced installation; however, one shall provide the possibility to take one of VSD out of service without affecting performance of other devices. Panel cooling is provided either in the form of overflow central fan or application of independent VSD fans.

Regulation of drives is in general similar to standard lay-out of MCC with hardwired signals of rate

control / feedback with PCS / DCS via interface panel of 4-20 mA.

Regulation user nodes.

As a rule process motors are automatically controlled and each motor has a regulation user node consisting of “start, stop” switch, “hand-off-automatic” selector switch (MANUAL / OFF / AUTO) and “start, stop” indicating lamps, which are installed near the motor. A-meters showing motor current are provided in regulation user nodes for medium voltage motors CH. A-meters are connected to secondary current 1A, remote A-meters are installed in MCC.

For motors which are automatically controlled with PCS / DCS, signals are transmitted between logical regulation mechanism at the plant and connection interface within SRR observational satellite system through patch panels and fiber-optic cables. Control organs include but are not limited to:

- Motor start
- Motor stop
- Motor status (operates)
- Motor drive error
- Motor in automatic selection mode (available for start)

Motors without regulation user node installed near motors (such as fan electric motor, etc.) also have hard-wired emergency stop button connected directly to motor control circuit.

Control stations are installed as a rule on a corresponding steel support near to the motor the control and are connected directly to motor start in control circuit.

10.3.6.4.2. SWITCHBOARD 10 KV

Medium voltage auxiliary switchboard PYCH 10 kV consists of two-section distributing panel installed in turbine house electrical room, $I_H = 1250A$, $S_H = 8371.05$ kVA.

Structure of PYCH 10 kV switches:

Switch with gas isolation GIS for $U_H = 10$ kV has double busbar according to IEC standards

Protection relays are based on programmed microprocessors and are communication objects. Transformers of voltage and current of certain level, output class and accuracy for protection and measurement are given in corresponding documents.

All circuit breakers within assembled switch are installed as a single bundle 10 kV.

10.3.6.4.3. LOW VOLTAGE TRANSFORMERS

Low voltage transformers 10/0.4 kV, 10/0.69 kV, 0.69/0.4 kV are provided in dry running with the capacity of 100, 200, 500 and 2500 kVA.

Transformers are installed in low voltage switchboard electrical room.

10.3.6.4.4. LOW VOLTAGE SWITCHBOARDS

Low voltage switchboards:

- 690V for motors 400 kW;
- 400V for general application and motors.

Low voltage bus ducts.

On the low part of transformer (with secondary current above 1600A) application of low voltage bus ducts is provided.

In this case metal sheath of bus ducts has IP56 index of protection

Bus duct is fixed at a safe level.

Instruments and controls power supply.

Systems of one phase uninterruptible power supply UPS with redundant duplication 230 V are

provided for devices energy supply as well as in the main control room. Systems are provided to control the process / distribution systems and energy control (PCS / DCS) with 30 minutes standalone downtime of battery operation with full rated load, and as for fire and gas system – with a battery with 24-hour standalone downtime at full load. Uninterruptible power supplies (UPS) have minimum 20% from the total idle capacity for development of measurement and control systems.

Every UPS system includes:

- Double rectifier / charger, each one is rated for 100% of load + battery charge;
- Double battery, each one with a capacity of 50% of the load for complete standalone downtime;
- Battery isolating circuit breakers (switchers);
- Double inverters, each one is rated for 100% of the load;
- Potential gearshift selectors;
- Support by-pass commutation switch;
- By-pass transformer;
- Main fuses of the header for devices distribution panel supply;
- Control room for panels located in SRR electric room or in main control room for deployment of basic system components.

Batteries are located in a separate room at rackings with protection against acid spills and with redundant exhaust ventilation system directed from the building. Eyes flushing tools and / or safety shower are provided in these rooms. Each room has a separate door leading outside.

Rectifier/charger energize from a separate MCC (SRR) or individual system of low voltage collecting bars HH. If original plant is equipped with emergency power supply, one of rectifiers / chargers energizes from this source. Bypass supply can be received from any of the above mentioned sources.

Calculation of main circuit power load flow see in appendix 2

10.3.6.5 DIRECT CURRENT UNITS

Every plant contains one battery system 100 Vdc to satisfy the needs of energy supply to distributor control and protection of equipment and other devices such as plant annunciators. Systems are designed for equipment inside the plant, with a battery for 4 hours standalone downtime with full load. The system has at least 20% from the total idle capacity for plant development.

Every direct current system has:

- Two chargers with the capacity of 100% of the load each + battery charger;
- Double batteries with the capacity of 50% of the load each for the whole standalone downtime;
- Battery isolating circuit breakers (switchers);
- Blocking diodes of rated load;
- Direct current panel with circuit breakers for complete supply from buses, deenergizing from buses, signaling device supply, etc.;
- Battery control panels.

Chargers have a possibility of load distribution.

Batteries take from the system up to 90% of their capacity within 8 hours. Battery life cycle in normal operation as well as with the specified recharging cycle is not less than 15 years.

Chargers energize from individual MCC or from an individual section of low voltage busbar HH. If emergency power source is available at original plants, one of the chargers energizes from this source.

10.3.6.6 PROTECTION RELAY

Protection relay is provided in volume corresponding to the existing «Electrical installations code» and «Protection relay guidance».

To protect electric equipment from damage the project provides the following protection devices:

- for local switch: protection against minimum voltage, protection against overcurrent, protection against ground fault current, maximum current protection, protection against current overload and ground short-circuit, operation workout relay, data transmission device, etc.;
- for line transformer: transformer temperature control, protection against overcurrent, protection against ground fault current, maximum current protection, protection against current overload and ground short-circuit, operation workout relay, data transmission device, etc.;
- for busbar coupler: synchronism check relay, protection against overcurrent, maximum current protection, operation workout relay, data transmission device, etc.;
- for electric generators in accordance with the set of manufacturing plant;
- for electric motor: protection against minimum voltage, thermal protection against overcurrent, instantaneous protection from overcurrent, protection against ground fault current, operation workout relay, data transmission device, etc.;

Protection relay is provided on microprocessor devices.

Circuits of control, automatic equipment, alarm and protection operate from DC control power =110V.

10.3.6.7 COMMUNICATION AND ALARM

For internal communication at the power plant the project provides effective and common communication and alarm devices.

The project provides the following kinds of effective and common communication and alarm:

- effective two-way telephone communication;
- loud-speaking command paging;
- common telephone communication (CTC);
- radio broadcast network of social facilities;
- electrical master clock;
- guard telephone communication.

Telecommunication systems of electric power plant shall be as follows:

- computer and telephone network
- radio communication system
- PA/GA system
- CCTV process operation automatic control system

Structured cable system shall be installed for expansion of LAN (local area network) and VOIP telephone lines at the power plant. It shall provide basic cabling to support various industrial environment and various suppliers environment. Structured cable system shall use cables U/UTPC at 6 for buildings horizontal cabling which are able to support 250 MHz of frequency and up to 10 Gbit of data up to 100 meters. Compatible RJ45 outputs (No. 2 of every point) category 6 shall be provided within the working area inside buildings.

Telecommunications cabinet shall have a possibility to locate network equipment, UTP (unshielded twisted pair) control and fiber-optic cables using corresponding distribution panel.

Distributing board cabinet of electric plant building shall be connected with the help of fiber-optic cables LAN and VOIP of PABX technical park network.

Two individual rings shall end inside every BD of electric power plant:

- Ring 1 (100 Gbs/sec for connection of all buildings, storage facilities and power plant)
- Ring 2 (100 Gbs/sec for connection of all buildings, storage facilities and power plant)

VOIP telephones shall be located in power plant buildings. They shall be protected against bad weather (IP54 minimum) in industrial buildings of wall-mounted and tabletop closed type in control room and in offices.

VOIP telephones shall be controlled with the help of VOIPPABX (private ATS with outcoming and incoming communication).

System of plant radio communication of technical park shall also cover electric power plant. All radio equipment shall be able to perform duplex communication using UHF frequency range.

Control room of power plant shall be equipped with additional communicator console. It shall be tabletop type and equipped with hand microphone. Besides there shall be a certain number of portable hand radio stations including those with removable and rechargeable large-capacity batteries and various accessories (units multi-charging devices, carrying cases and belts).

Power plant shall be equipped with additional PA/GA node of the plant PA/GA system. It shall be able to transmit speech and alarm signals to corresponding loud-speakers distributed over the power plant. Level of loud-speakers sound shall be not less than 6 dB, louder than the level of external noise in order to provide correct speech and legibility of signals. In the environment where noise level exceeds 84 dBA visual alarm signals (beacons) shall be provided in addition to loud-speakers broadcasting.

PA/GA system shall be coupled with F&G (Fire&Gas) system in order to transmit alarm signals via loud-speakers and to control coupled radio beacons in external areas and buildings of the power plant.

Additional PA/GA access panel shall be located in the power plant control room. It shall be able to transmit instant speech, emergency speech and perform activation of alarm signal via power plant loud-speakers.

External areas and industrial buildings of the power plant shall be equipped with weather-proof horn loud-speakers and radio beacons (IP65 minimum).

Internal system of paging shall provide voice communication between operator in power plant control room and field office located at the power plant.

Field offices shall be equipped with acoustically protected cab, source of audio signals and radio beacons.

Operator's console in power plant control room shall have two-way communication with internal plant paging system located in the main board of IGCC.

Electrical master clock of the power plant is provided by installation of clock microprocessor station STS type at the communication nod. Recommended configuration of the station is rated at connection of four independent secondary impulse clock control stubs controlled by heteropolar minute pulses.

Equipment for power plant access control consisting of controlled doors in main rooms requiring security access. Doors for personnel access are foreseeingly controlled while all others are blocked.

In case of emergency all controlled doors shall be automatically unblocked in order to enable evacuation of the personnel.

Controlled doors consist of the following:

- Door controller: provides connection and control of door access equipment (proximity smart card readers, magnetic latch and door release buttons).

- Proximity smart card readers: shall be located near the door entrance. They shall read cards at the distance of not less than 40 mm and include LED indication to reject the access or provision.

- Magnetic latch: consists of two parts: electric magnet and reinforced plate.

- Door release button: door exit button is located inside the room and is used to deenergize magnetic latch and allow personnel to go out.

- Emergency door release button: is used when the controller fails to open the door and someone turns out to be in a trap inside the building.

CCTV system shall be controlled via LAN network as well as via video equipment of the power plant. Cameras shall be PTZ type (Pan/Tilt/Zoom), designed for work in an industrial environment (weather resistance IP65) and shall provide high definition and brightness pictures from the areas in normal daylight or at night with plant lighting. Cameras shall be displayed at PCS (process control system) of the console as well as on a big screen located inside the power plant control room.

CCTV system at the power plant consists of several CCTV cameras, located at the key points in order to have appropriate observation in this area: external sites around the main building, distribution section, control room.

10.3.6.8 LAYOUT OF CONFIGURATION AND CONSTRUCTION OF LOW AND MEDIUM VOLTAGE ELECTRIC DEVICES.

Plants with elevated cable constructions with external transformers. Along the edge of transformers there is a possibility of rolling the transformers to the place of repair. Plant is a concrete structure.

Transformers, switches, busbars, batteries, chargers are located according to engineering practice and comply with the GOST IEC 60519-1-2011 "Safety in electroheat equipment. Part 1. General requirements". Sufficient height is provided in cable cellar (minimum height from upper beams to the floor)

Plant building is located in non-hazardous area. All plants shall have fire resistant walls, doors and reinforcement. Stability testing shall last 2 hours.

Each plant has a transformer installed on its stretched side upgraded with intermediate fire protection for transformers, space for repair and external fencing in order to avoid outside interference.

Standard plant includes but is not limited to the following basic equipment:

- MV/MV Transformers with individual neutral grounding resistors;
- MV/LV with busbar transformers;
- Air-conditioned buildings with electric switch room, communication and gas room;
- MV internal GIS (gas isolation switch);
- MV internal "logical" motor control centers (MCC);
- LV internal switches with double metal twisting;
- LV internal "logical" motor control centers (MCC);
- Chargers and batteries for 110 V dc system;
- Rectifier/charger, batteries and invertors for electric system 230 V ac;
- Equipment for observation and energy supply;
- Failures and activity sequence logger;
- Digital failures logger;
- Individual protection panels, PMCS and "logical" control centers;
- Fire alarm panel;
- Panel of heat, ventilation and air-conditioning system (HVAC).

Plant designed above the level of grade level with elevated cable cellar. Transformer yard with also be at the same level with adjacent fire-resistant wall extended to the level of the first floor. The plant has no windows, but it has a door for personnel and one door for equipment in various parts of the building. Door aperture for equipment will be enough for entering the biggest assembled parts of equipment as well as it will be equipped with standard doors with access panels, if necessary.

Underground supply cables are laid below the ground and are directed through devices to power plant transformer yard. Busbars from transformers enter the power plant through joining wall above the installed switchers.

Energy supply and control cables for equipment and operation areas exit the power plant through fire resistant exits located on the floor and are supported by cable holders in the elevated cellar zone. Internal cable can be installed above the panels and, where necessary, can exit the building through fire resistant exits in the wall or to cable cellar.

Power plant location has space for extension of switch and control arm as it is described above. It is supposed that this space is enough for additional set of equipment for each bus section.

Minimum vertical gap 450 mm above the equipment if measuring the distance between the highest part of the equipment and the bottom of ceiling beam, air conditioners exits or output supporting cable systems. Building floor is even and smooth in order to facilitate the rolling of switches installed at one level with the main floor.

Internal part of the building has positive pressure (heating, ventilation and air conditioning) of HVAC system in order to provide exit of dust and adverse gases outside. Batteries are located on the shelves in a special room protected against acid and equipped with additional ventilation system leading from the building. These rooms have all conditions for shower and eyes flushing in case of an accident. All rooms with batteries are classified as zone 1 of IIC gas group in order to consider availability of hydrogen.

10.3.6.9 INTEGRATED CABLE SYSTEM

Cables 35 kV, 10 kV and low voltage (HH) are rated at:

- Short circuit thermal power;
- Loss of voltage;
- Continuous current rating.

All cables of primary systems and subsystems are designed for continuous work. All cable dimensions consider temperature of the environment / soil, soil heat resistance and group thermal conduction

Cable dimensions are able to withstand the loads of short circuit current similar to values of distribution device and time of elimination of protection devices short circuit.

Cable 35 kV.

- Isolation made of sewed reinforced polyethylene or polyvinylchloride with flame retardant PVC cover.
- Single core, brass: dimension 300 mm², or according to calculation.
- Applicable standards: GOST IEC 60332-3-21-2011 “Tests on electrical and optic cables under fire conditions. Part 3-21. Flame spread of vertically-mounted bunched wires or cables”.

Cable 10 kV.

- Isolation made of sewed reinforced polyethylene or polyvinylchloride with flame retardant PVC cover.
- Single core, brass for feeders: according to calculation of cable dimension.
- Multicore, brass for motors: according to calculations.

- Low voltage cables HH (690 V and 400 V)
- Isolation made of sewed non-reinforced polyethylene or polyvinylchloride with flame retardant PVC cover.
- Multicore or single core: minimum dimensions: 2.5 mm² for energy supply and 1.5 mm² for control.

In general cabling will be made:

- From closed type main switchgear 3ГРП-35 kV to reducing transformer 35 kV/10kV – below the ground in cable duct covered up with sand;
- From 3ПВ-10 kV to reducing transformers of processing units 10 kV/0.66 kV/0.4 kV – below the ground;
- Cables 10 kV within power unit for feeder – below the ground;
- Low voltage cables: above the ground, as high as possible, in metal cable trays.

All CH cables, buried directly in trenches, will be laid in one layer. Cable shall be laid on clean compacted sand 75 mm thick with cable protection tiles, laid on the top. Trench shall be filled to the soil level. Cable protection tiles are not required at the plant sites with solid surface. If cables are laid in two layers, sand filling between the layers shall be 150 mm.

All motor control cables shall be installed together with corresponding power supply cable.

Cable coming out of the ground for connection to the equipment shall be laid in a pipe resistant to UV not less than 150 mm above the ground. Pipes shall be pressurized after cable lying in order to protect it against environmental impact.

Cables crossing roads shall be laid in concrete polyethylene pipes and equipped with draw rope. At least 20% of spare pipes but not less than two, pressurized from both sides, shall be installed for further needs.

Cable tray or ladder-type trays supported by structures, shall be, as a rule, used for several overhead cable passages

Cabling into cable tray will be performed in one layer with the distance for CH cables, one layer for low voltage cables and in two layers for control cables.

Cable tray covers shall be equipped with mechanical protection (on the upper layer for horizontal and vertical passages). Cable trays and covers for energy supply and control shall be rated at heavy work and shall be hot galvanized after manufacturing.

All cables included in any part of electrical equipment or unit shall be compacting with the help of double brass cable glands. All cable inputs shall have earth tag and PVC protection tie. All cable inputs which will be used in areas 1 and 2 shall be explosion proof.

Cable numbers shall be applied to cables along the routes and on both connection points. For underground cabling the distance between the numbers along the route shall not exceed 5 m, and for overhead cables – 25 meters. Cables shall be numbers in places where they divert from the basic route.

For underground cables of unpaved areas overhead markers of cable passage shall be made every 25 m, as well as upon every route directional change and on both sides of road of pipeline intersection.

Enough space (minimum 600 mm) will be maintained between energy supply cables and signal cables of instruments and controls in order to avoid strays.

Power supply cables and control cables shall come to the equipment and working areas from the power plant through fire resistant exits located on the floor and shall be supported by cable holders in the elevated cellar area. Internal cable can be installed over the panels and, where necessary, come out of the building through fire resistant exits in the wall or into the cable cellar.

All field equipment with tapped inputs shall be suitable for metric thread.

10.3.6.10 GROUNDING DEVICES, LIGHTNING PROTECTION

Grounding system ensures:

- Personnel safety
- Static current protection
- Lightning discharge protection
- Electromagnetic interference protection.

Grounding devices and lightning protection are provided based on the Electrical Installation Code of the RK, Instruction on installation of lightning protection device for buildings and structures (SN RK 2.04-29-2005), Manual for Instruction on installation of lightning protection device for buildings and structures (RD 34.21.122-87) and Guidelines for protection of 3÷500 kV power stations and substations against direct lightning strikes and storm waves from power lines (Moscow, ORGRES, 1975). Resistance of the grounding device any time of the year shall not exceed 0.5 Ohm.

The grounding system is mounted around the substations, processing units, structures and other electrical facilities. All grounding systems are connected in two or more points to make a shared network. Equipment which shall be grounded includes reservoirs, vessels, carriage, rack, motors and other electrical equipment. All metal non-current-carrying parts of electrical equipment, which can be under voltage, shall be connected with the grounding system. If there is no the grounding system nearby, equipment shall be connected to one or more special grounding pivots.

Bare copper main conductor of the grounding network has section of 70 mm², except for substations which bare copper conductors shall be 25mmx3mm.

The grounding system covers the whole facility in terms of the main earth tag with interconnection to equipment and structures, which shall be grounded. Metal housings of electrical equipment are connected with the earth tag of the power station. The metal housings of non-electrical equipment, for example, vessels, are also connected with the earth tag of the plant.

Underground interconnection of earthing grid is performed by thermic welding. Aboveground interconnection of earthing grid is performed by compression connectors, and connection of the grounding to equipment / structures shall be performed by bolting.

Lightning protection and static grounding protection is provided by connecting the steel structures, towers, vessels, reservoirs, etc. directly to the grounding system. High steel structures, such as towers or structural columns, if they are electrically continuous, are virtually considered to be protected against lightning because of their connection to the grounding system. Connections at the junctions can be used to ensure electric continuity if required.

Electric continuous supporting columns can be used as down conductors that help to connect the reservoirs, vessels, etc. to the grounding system.

Flanges of metal conduits, which have insulated inserts, shall be interconnected to ensure electric continuity. Flange joints without inserts or continuously welded c are deemed by their nature interconnected. Pipelines shall be connected to the grounding system at points where they are within and beyond a battery or in case of entering or exiting the danger areas.

Lightning surge protection of GTPP, SS, HVL and other structures is performed as follows:

- against direct lightning strikes – by lightning rod and earth wire;
- against waves incoming from outgoing lines – by lightning rods and at a defined length of these lines by protective devices installed on the ways and in switching equipment which include valve-type arresters (VTA), excess-voltage suppressors (EVS), tubular arresters (TA) and protective spark gaps (PSG).

Lighting protection of buildings is performed as per RK 2.04-29-2005 “Instruction on lighting protection of building and structures”. Buildings and site facilities are protected by category I of lighting protection where there are explosive zones of V-1; V-1a; V-1g type as per the Electrical Installation Code. Lighting protection is performed by using lightning rods, Satellite-type lightning rods to be mounted on floodlight towers or on special towers, and by using lightning rod grid (faraday's cage) placed on the building roofs.

All explosive zones are covered by lightning rod protected zones.

Protection against external high voltage is arranged by grounding the metal suspended pipelines and installing equipotential sealers between parallel industrial pipelines.

Lightning protection system is integrated with the grounding sources and external grounding grid. Its resistance shall not exceed 0.5 Ohm (at any time of the year). Lighting protection includes lightning rods, diverters and ground wires.

Potential equalization is provided for all metal structures as per the Electrical Installation Code. For potential equalization, the technical rooms have the mounted equipotential bars (EPB) which are connected with foundational ground wire via builder's work projectures of concrete structures of the building.

Network taps have minimal cut set, however when selecting the cut set it is necessary to measure voltage to ground not more than 50V at maximum short-circuit to ground.

Grounding shall be provided for all metal parts of electrical installations and other metal structures which are normally not under voltage, but which can be under voltage, as well as metal conduits, air ducts, cable structures and switchboard cabinets. All metal non-current-carrying parts of electrical equipment and exposed conductive parts of lamps shall be grounded neutral by connecting to the neutral protective conductor laid from PE bar of lead-in distributor. For neutral grounding, 3 and 5 conductors of supply and distributing networks are used. Metal housings of shower trays, metal housings of washers shall be grounded neutral. For neutral grounding, PV1 conductor with section of 4mm² is used which is laid from PE bar of floor distribution board and is hidden in polyethylene pipe with diameter of 20 mm, in floor preparation. The equipotential bonding system shall be performed at the facility inlet. For this purpose, metal pipes of utilities, air ducts, cable structures and protective conductors of electric main join in with the Main Grounding Bus inside the lead-in distributor in switchboard room. For protection of service staff against electric shock, all normally non-current-carrying parts of electrical equipment which can be under voltage in case of insulation failure shall be grounded neutral by connecting to neutral protective conductor of the main as per the Electrical Installation Code. The Main Grounding Bus is joined with external grounding grid.

10.3.6.11 ELECTRIC LIGHTING, WELDING NETWORK

Low-power (400/230 ACV) system of lighting and power distribution, 3P + N + PE, is provided for substations, SRR, and other large buildings around the facilities. Additionally, the external panels shall be located around the facilities to provide the local feeders on remote sites. The whole lighting and low power circuits shall be protected by corresponding circuit breakers.

Lighting fitting is fed from lighting distribution boards which have automatic circuit breakers for protection of each subscheme and enable a separate insulation of the circuit. The lighting fitting is designed for operation from 230V monophasic network. The lighting distribution boards consist of 4 three-phase wires and, if applicable, are located in safe place, preferably within the building.

The use of high-pressure mercury lamps shall be maximum as far as practicable to light the buildings and operating platforms, lower pipe racks, etc. across the whole plant. However, the luminescent lamps are provided in strategic places and for emergency lighting.

The lighting fitting for general lighting shall be located as close to control and measurement instruments, etc. as possible to avoid special lighting for these devices.

All exterior lighting of the power station and highlighting shall be controlled by electrical photocells. The use of local circuit breakers shall be minimized. The steam sodium high-pressure lamp shall be used for the whole exterior lighting and flood lighting.

The lighting distribution boards shall be located within the area of service. All outgoing schemes in the boards located within the plant shall be controlled by double-pole miniature circuit breakers.

The lighting fitting specially installed for lighting the sight glass shall be connected according to the separate subscheme and controlled by locally installed circuit breaker.

Aircraft warning lights shall be installed on all structures at height of 45m and higher as per Annex 14 ICAO and local standard. The lamps used for aircraft warning lights shall have long service life (at least 20,000 hours).

Material for lighting systems/ sockets shall be one of the following:

- GRP (glass-reinforced polyester)
- HD galvanizing steel
- Copper-free aluminium alloy (for danger areas)

All lighting and low power equipment which is installed outdoors has IP65 protection level, being explosion-proof equipment.

All lamps and sockets shall be duly certified for an area where they are installed (as a rule, explosion-protected electrical equipment / explosion-proof motors), and have the corresponding documentation.

Small power supply schemes shall be designed for ensuring an easy access to the use of mobile or hand-held equipment around the facilities. Single-phase power supply shall be distributed around the buildings and production premises, while three-phase power supply shall be distributed across the whole process and external zones only.

All sockets shall be standardized to ensure that only type of socket is required in the system, and can be used with all sockets installed within the plant. External plugs and welding sockets meet the IEC standard.

Sockets which shall be installed in places with potentially higher humidity shall be protected by circuit-breaker which has opening of circuit by shorting to ground.

Level of lighting meets SNiP RK 2.04-05-2002 “Natural and artificial lighting”, manual to SNiP II-4-79 and BCN 34-91.

Welding network is made as per the guidelines on performing the welding network on thermal electric power stations and BCN 22-80.

10.3.6.12 HEAT TRACING

It is expected that inclement weather in winter and lack of evaporation capacity around the facilities will result in necessity of vast network of electric circuits of heating.

Local 3P+N switchboards are provided in safe areas which feed self-regulated belt-type schemes and temperature-controlled lines.

Control system is electronic, has status signals and data transferred to PMCS system via substation units.

Section 10.3.7 ARCHITECTURAL-BUILDING SOLUTIONS.

Basic designs for architectural-building solutions of the building and structures of the gas turbine station site are given in Volume 1, book 2, section 8.

Section 10.3.8. HEATING, VENTILATION AND CONDITIONING.

10.3.8.1 Weather conditions of construction site.

Climate of the area is extremely continental. Climatic features of the construction area in Atyrau are given in table 10.3.7.

Table 10.3.7 - Climatic features of the construction area

Item	Indicator name	Value	Justification
1	Climatic area	IB	SNiP RK 2.04-01-2010 Pic.1, p.108
2	Design outside temperature, °C		
	Absolute minimum	-38 °C	-«-table 1, p.60
	Absolute maximum	+43 °C	-«-table 2, p.78
	Coldest 24-hour period	-30 °C	-«-table 1, p.60
	Coldest five-day period	-25 °C	-«-table 1, p.60
3	For heating system engineering	-25 °C	-«-table 1, p.60
4	For ventilation system engineering		
	In warm period	+31,2 °C	-«-table 2, p.78
	In cold period	-25 °C	-«-table 1, p.60
5	For conditioning system engineering	+34,2 °C	-«-table 2, p.78
6	Average air temperature of a period with mean daily temperature ≤ 8 °C	-3,4	-«-table 1, p.60
7	Duration of heating season, 24-hour period	177	-«-table 1, p.60
8	Mean monthly relative air humidity, %		
	Coldest month	84	-«-table 1, p.60
	Warmest month	46	-«-table 2, p.78
9	Barometric pressure, Kpa	1020	-«-table 2, p.78

Heat conductor for heating and ventilating systems is low pressure steam with parameters of the heat conductor $T = +130$ °C; $P = 5$ bar.

10.3.8.2 JUSTIFICATION OF ACCEPTED SYSTEMS AND CRITICAL DECISIONS

Estimated coefficients of thermal resistance of enclosing structures are calculated for all heated buildings.

Air temperature in premises of the building and strictures is accepted as required by normative documents, technical requirements of manufacturing plants on operation of equipment, systems, devices and instruments. Inside temperature and humidity which is maintained by the heating and ventilating systems is specified on the HVAC drawings.

When calculating capacity of the heating systems of the buildings, the following is taken into account:

- 1) heat losses through the enclosing structures;
- 2) heat consumption for outdoor air heating;
- 3) heat losses for materials, equipment and means of transport;
- 4) heat emission from power and lighting equipment, production equipment, communications, for, people and other sources.

The project provides the air-type heating system to ensure the permissive environmental conditions and air purity in working area of the premises.

Heating pipelines in places of crossing of overlaps, interior walls and compartments are laid in non-combustible material shells; shell edges are aligned at the same level with surfaces of walls and compartments, and by 30mm higher than the mark of finish floor of compartments. Gaps and apertures in places of pipe laying are filled-in by non-combustible materials.

Heating pipelines are made of steel pipes suitable for water and gas under GOST 3262-75 and steel electric-welded pipes under GOST 10704-91. Anticorrosive coating of steel pipelines is GF-021 primer (1 coat) under GOST 25129-82. Above the primer, the steel pipelines are coated by MA-025 oil-base paint in 2 coats under GOST 8292-85.

To ensure the air conditions set by sanitary standards and production requirements in, the premises provide:

- combined extract-and-input systems of ventilation and conditioning with standardized ventilation rate;
- special set of measures on noise control and vibration reduction when operation of ventilating equipment;
- automatic control of ventilation device operation.

The project provides installation of combined extract-and-input system with sections of heat recirculation (recuperation), and separate systems of forced inlet and exhaust ventilation. Air is distributed in premises by air ducts, and air is supplied to the premises by diffusers or through ventilating grills.

Air handling units with recuperators are completed with inlet and exhaust fans, inlet and exhaust filters, recuperator (recirculation section), heater, cooler, steam humidifier and automatic control system with remote-control desk. Fans of the air handling unit are equipped with high-performance impellers with forward bent blades and asynchronous motors. Sealed ball engine bearings do not require routine maintenance and do extend service life. Fan motors are protected by thermal contacts with automatic restart at 125°C, built in equipment.

The project provides a deployment of filters of first degree to purify inlet air from dust and air heaters for additional purification – prior to supply air to the premise.

See payments and air exchanges accepted in the project as required by Annex 10 SNiP RK 4.02-42-2006 in a list of the documents attached to the drawings.

The project provides the smoke control systems of a building as required by section 9 SNiP RK 4.02.42-2006. Smoke exhaust system works together with the fire alarm system. Upon the fire alarm system signal, the smoke exhaust fans and smoke valves of smoke exhaust system are activated. The smoke exhaust fans are installed to remove the combustion products appeared in case of a fire with temperature 400-600°C in the smoke control systems of the buildings, except for A and B category premises. For A and B category premises, the explosion-proof fans are installed. The explosion-proof fans meet the requirements of safety and health at the enterprises with explosion-dangerous production. These fans provide the measures and means which eliminate the possibility of sparking while in operation.

10.3.8.3 BUILDING OF ELECTRIC DISTRIBUTION SUBSTATIONS

Designed conditioning system maintains the required conditions of inside air and seals using the HVAC system as described below to meet the requirements of SNiP RK 4.02.42-2006.

Cable spreading room

Ventilation system of the cable spreading room consists of inlet and exhaust fans on opposite angles of the room to maintain the necessary temperature as required by SNiP RK 4.02.42-2006.

The smoke exhaust system reliably uses the same ventilation system, therefore the inlet fans, exhaust fans and corresponding accessories are used at high temperatures (> 400 ° C).

Electric distribution substations (second floor)

Ventilation and air conditioning system is fully air, of channel type, with constant air volume, with several heating areas, using outdoor air and discharging air from the premises to the outside.

The mechanical supply system is equipped with two ventilation units (working and standby) with hot water calorifiers.

These units have one common inlet of outdoor air equipped with motorized gas damper. Because of adversative environment, the fresh air intake provides preheating.

To control and maintain the designed humidity, the main supply air duct has one steam humidifier installed.

To cool wet-type air coolers (calorifiers), the water-cooling system is installed in the air conditioning system at electric and distribution substation. The system is equipped with two water cooling units (working and standby) and two pumps with cooled water (working and standby). Units and pumps are located in the HVAC premise – at the first floor.

Air-conditioners, fresh air receiver, preheating unit, humidifier with power board / automatic control board are located in the HVAC premise – at the second floor.

Power board / automatic control board shall feed and control all system equipment.

Premise for distribution device shall be protected against fire by the gas suppression system, and this premise has the ventilation and smoke exhaust system.

Each ventilation system consists of proper quantity of exhaust fans. Central air-conditioners provide the required quantity of treated air.

Air pressurization system for evacuation routes consists of the required quantity of inlet fans to avoid ingress of smoke on the stairs and to a corridor.

Battery room

Battery room has a separate waste air ventilation system with two exhaust fans (working and standby). The fans shall be certified for classification of danger zones (zone 2, IIC, and T1) of the battery room. Repeated air circulation from this room is unacceptable.

Air heaters of zone heating with heat tracing are provided, if required, to control temperature of separate conditioned zones. The same air heaters of zone heating serve for maintaining the required temperature conditions to ensure the technical characteristics as required by SNiP RK 4.02.42-2006.

During the stage of completion of device operation, the conditioned zones are not provided with the separate heating system.

The smoke exhaust system is serviced by the same exhaust fans of the battery room.

10.3.8.4 SWITCHBOARD BUILDING

The building is serviced by the HVAC systems, as described below, so as to meet the indoor conditions as per SNiP RK 4.02.42-2006.

Ventilation and air conditioning system is fully air, of channel type, with constant air volume, with several heating areas, with supply of outdoor air and discharge of air to the outside.

The ventilation system is equipped with two ventilation units (working and standby), the heat supply system of calorifiers is water system; a section for air cooling in warm period of a year is stipulated.

These ventilation units have one common inlet of outdoor air equipped with motorized gas damper and explosive valves. Because of adversative environment, the fresh air intake provides preheating.

To control and maintain the designed humidity, the main supply air duct has one steam humidifier installed.

To cool the conditioning air coolers (calorifiers), the water-cooling system is installed at electric and distribution substation. The system is equipped with two water cooling units (working and standby) and two pumps with cooled water (working and standby).

Air-conditioners, fresh air receiver, preheating unit, humidifier are located in the HVAC premise.

Power board / automatic control board shall be located in the HVAC premise or in electric room and feed and control all system equipment.

The ventilation systems for elimination of smoke and gas shall be in raised floor of the premise of the distributed control system, in raised floor of the engineering premise, in floor of the offices, manual control premises, in telecommunications closet, in floor of the server room protected against fire by the gas extinguishing system.

The ventilation system consists of air handling units and appropriate exhaust fan. The channel-type system with constant air volume, using outdoor air and discharging air to the outside.

Air pressurization system for evacuation routes consists of the required quantity of inlet fans to avoid ingress of smoke on the stairs and to a corridor.

Toilet rooms and WC facilities are equipped with separate system of mechanical exhaust ventilation with one exhaust fan. Air from toilet rooms is discharged to the outside.

Electric-type air heaters of zone heating are provided, if required, to control temperature of separate conditioned zones. The same air heaters of zone heating serve for maintaining the required temperature conditions during the stage of completion of device operation.

During the stage of completion of device operation, the conditioned zones are not provided with the separate heating system.

10.3.8.5 TURBINE HOUSE

The building is equipped with the ventilation system to control maximum temperature inside the premise in summer-time. The separate heating system maintains minimum temperature inside the premise in winter-time.

The turbine house is ventilated fully by the air conditioning system, of one channel type with constant air volume, with 100% fresh air supply, equipped with inlet fans located inside the HVAC premise of the substation, at the second floor. Fresh air is supplied through the grid with pressure release controls located at the roof to ensure through flow ventilation of the building. This system (inlet fans and exhaust grid with pressure release controls) shall be used in case of a fire to clean from smoke, therefore the grids and pressure release controls shall be certified as per the Kazakhstani standards as the smoke exhaust devices.

In winter, the premise ventilation is minimized up to $6 \text{ m}^3/\text{h}/\text{m}^2$, therefore the heating system shall be implemented through the heating element with air duct fed by low pressure steam, installed only on one inlet fan.

In winter, an additional heating system is provided which ensures an appropriate quantity of electric heating elements to maintain $\geq 5 \text{ }^\circ\text{C}$ temperature inside the premise during the completion of the device operation.

This building is not a building type for designing of blast.

10.3.8.6 BACK OFFICE LOCATIONS

The back office locations are equipped with the heating system which maintains minimum temperature inside the premise in winter.

The heating system is implemented through an appropriate quantity of heating elements fed by low pressure steam. The heaters are equipped with fan, heating calorifier and fresh air intake.

Additional heaters fed by low pressure steam are installed to increase a ventilation ratio of the premise from 5 to 8 for the purpose of the premise emergency ventilation.

In winter, an additional heating system is provided through an appropriate quantity of electric heating elements to maintain ≥ 5 °C temperature inside the premise during the completion of the main system operation.

Outer walls and roof of the premise for the diesel fuel pumping station are not explosion-proof type of the project.

10.3.8.7 REQUIREMENTS TO EQUIPMENT. FEATURES OF AIR DUCT AND PIPELINE MATERIAL

All equipment of the heating, ventilation and conditioning systems shall be certified and meet the stated codes and standards.

All equipment of the heating, ventilation and conditioning systems is up-to-date and one of the latest models from the approved vendors list below, see Table 10.3.8.

Table 10.3.8 – Heating system equipment

Name of equipment	Manufacturer
Conditioning units	Carrier Trane York
Chillers/condensing units	Carrier Trane York
Blast valves	Ruskin Andair

The alternative equipment will be selected at the stage of engineering documentation.

Types of coolants have been selected for work in severe climatic conditions and taking into account the Montreal Protocol.

Power supply of equipment of the heating, ventilation and conditioning system with motors with capacity of less than 0.18 kW is accepted by one-phase 220 Volt/50 Hz, equipment of the heating, ventilation and conditioning system with capacity of more than 0.18 kW is accepted by three-phase 380 Volt/50 Hz

All inlet ventilating equipment is located in ventilation plant rooms as required by section 8.10 SNiP RK 4.02.42-2006.

Air ducts of the ventilation system are expected to be made of the galvanized steel under GOST 14918-80* with sheet thickness as required by annex 11 SNiP RK 4.02.42-2006. External dimensions of the cross section of metal air ducts are accepted according to GOST 24751.

Air ducts laid outside the building shall be of 1 mm sheet thickness. The project provides P class air ducts. Combustible material air ducts are provided in D category single-store production premises, the remaining premises are equipped with non-combustible material air ducts. Fire resistance limit of air ducts is accepted as per table 2 SNiP RK 4.02.42-2006.

It is expected to install fire-stop valves in fire barriers and when crossing by the air duct the premises with different categories of fire danger.

Pipelines for heat supply and heating systems are made of metal as per GOST 3262-75 (GOST 10704-91) or of polymer materials.

Pipelines for cold supply systems are made of copper pipes. Condensate pipes are made of polymer materials.

Pipelines are laid in thermal insulation. Thermal insulation is provided as required by SN RK 4.02-03-2011 "Thermal insulation of equipment and pipelines". Thermal insulating materials with thermal conductivity of at least 0.05 W/m °C and thickness which ensures surface temperature of not more than 40 °C are used as thermal insulation.

For uninterrupted, fault-free and correct operation of all designed ventilation systems, ease of management and day-to-day control of ventilation device performance, the project provides maximum automation and blocking of ventilation devices and their remote control.

Section 10.3.9. WATER SUPPLY AND WATER DISCHARGE.

10.3.9.1 GENERAL DATA

Section of the water supply and water discharge systems of the gas turbine power plant of the National Industrial Petrochemical Technological Park has been developed based on the following documents:

- technical specifications for supply of water from Astrakhan – Mangyshlak water main;
- design tasks;
- site master plan;
- vertical planning

The project has been implemented as per the applicable codes and regulations:

- SNiP RK 4.01-02-2009 "Water supply. Public utilities";
- SNiP RK 4.01-41-2011* "Plumbing and sewage system";
- CH PK 4.01-03-2011 "Water discharge. Public utilities";
- SNiP RK 2.04-01-2010 "Building climatology";
- SN RK 3.01-01-2011 "Master plans of production plants";
- SNiP RK 2.02-05-2009 "Fire protection of buildings and structures";
- Standard for fixed water spray systems for fire protection" (NFPA-15, 2007).

10.3.9.2. WATER SUPPLY AND WATER DISCHARGE SYSTEMS

At the area of the gas turbine power plant, the separate systems of water supply and water discharge are designed which discharge duly domestic, production and storm waste waters of the corresponding waste discharge system, as well as the systems of utility and drinking water supply, fire-fighting main and circulation supply.

The water discharge systems are able to discharge properly the fire waters or storm waters which can be contaminated. All sewage runoffs are neutralized and cooled up to 45 °C or lower in their initial setting.

The following water supply and water discharge systems are provided:

- utility and drinking water supply;
- utility and drinking water supply for emergency shower;
- fire-fighting main;
- process water supply;
- supply main of circulation supply system;
- sewage system of pollution-free pure surface waste waters;
- sewage system of oil-polluted surface waste waters;
- waste water sewage system;
- oil runoff sewage system;

- domestic sewage.

10.3.9.2.1 WATER-SUPPLY SYSTEMS

To meet the technical, drinking and fire fighting demands for water of the gas turbine power plant, it is planned to use the designed general system of water supply of the technical park which includes:

- utility and drinking water supply;
- fire-fighting main.

Drinking water enters to the gas turbine power plant from the system of the utility and drinking water supply of the complex.

Utility and drinking water supply for emergency shower.

Emergency shower system consisting of emergency shower and eye washes is provided as the safety system for wash of human limbs which contacted with hazardous materials (acids or caustics) or in case of a fire.

Emergency shower system is designed for provision with drinking water as per the requirements to the drinking water for emergency showers and eye washes.

Emergency shower and eye wash are located on the following areas of the station:

- battery room (one eye wash only);
- area of a chemical dosing unit (one eye wash and one emergency shower);
- diesel fuel pumps and transfer pumps (one eye wash and one emergency shower).

Drinking water is supplied from the water pipe system of the complex and directed to the stinger ring of the emergency shower. All emergency showers and eye washes are parallel connected to the stringer ring to enable all equipment items to work separately.

Flow of the emergency shower and eye wash is controlled by an aperture for flow restriction to ensure the following flow:

- emergency shower: 4.5 m³/h;
- eye wash: 0.1 m³/h.

Diameter of stinger ring pipes is calculated taking into account two showering installations and one eye washes which work at the same time.

Minimum pressure which shall be on showering installations is 1 bar, etc.

Ground-surface feed pipelines have electrical heating to maintain water temperature >15°C. The drinking water system includes the supply and return pipelines to prevent water stagnation.

Fire-fighting main

At site of the gas turbine power plant, the fire water ring-main (Ø 400mm) is designed which is connected to the general firewater supply system of the complex by two lead-in wires.

Maximum water requirement of the HFW system is 589 m³/hour.

Minimum pressure is 8 bar.

Dictating water flow is accepted for fire extinguishing of the diesel storage tank.

Process water enters the gas turbine power plant from the system of process water supply of the complex.

The circulation supply system includes the open recycled water cooling system which uses water as a coolant, and is a water-supply system which works cyclically and consists of heat-exchange equipment, cooling units, water pumps, pipelines and other components.

Recycled water from processing units (Temperature: 45° C, Pressure: 5.0 bar gauge) enters the cooling tower for cooling. Cooling water (Temperature: 35°C) is collected in the reservoir under the

cooling tower. Water enters the suction mud tank through diversion channels. Thus, feed water enters the processing units for recycling.

As cooling water of the open recycled water cooling system absorbs heat, the carbon dioxide is discharged to air from the cooling system, turbidity value (nephelometric turbidity unit) and dissolved oxygen increases, content of dissolved salts increases, power fluid escapes. In this regard, quality of recycled water decreases. Deposits are formed, there is corrosion, clay is formed, growth of bacteria and algae increases resulting in loss of work of heat-exchange equipment and corrosion of the unit components. To solve these problems, it is necessary to control quality of recycled water, ensures that the parameter value do not exceed the set ones, it is necessary to add special scale inhibitor to the suction mud tank. Thus, formation of deposits will be decelerated. Concentration of the scale inhibitors shall be 5 mg/l (prior to specification). It is also necessary to add corrosion inhibitors to the suction mud tank resulting in slowing the corrosion process. Concentration of the corrosion inhibitors shall be 25 mg/l (prior to specification). Adding of bactericide agents to control the microorganism growth in the recycled water system will decrease the silt volumes. To perform abovementioned processes, it is necessary to install fully automatic systems of chemical agent pumping, and the adding process itself will be controlled by the automatic detection system which works real-time, the chemical agents will be added automatically to ensure the steady operation of the cooling water system.

Small volume of recycled water enters the measuring heat-exchange unit which is inside the premise to control water quality. This system uses steam as a source of heat. The device which controls water quality monitors formation of deposits and corrosion.

In the circulation supply system, the automated control system (ACS) controls water flow, temperature, pressure, oil temperature in a fan of the cooling tower, the vibration level and water level. The system measures on-site and centralizes control.

The cooling tower has reinforced framing, external decoration is made of reinforced plastic. Model of structure of the cooling tower resists single-point loads, non-symmetric loading, seismic and wind loads, other dynamic loads and decreases the vibration level. The project provides the measures on prevention of icing-up of the cooling tower structure in winter. The cooling tower is equipped with electrical satellites for ice melting and protection against sand.

Design capacity: 1453 m³/hour

Design parameters of water circulation:

Dry-bulb temperature: $\Theta = 40^{\circ}\text{C}$

Wet-bulb temperature: $t = 29^{\circ}\text{C}$

Feed water temperature: $t_2 = 35^{\circ}\text{C}$

Recycled water temperature: $t_1 = 45^{\circ}\text{C}$

Feed water pressure (around a border of the unit): $p_2 = 0.64 \text{ MPa}$ (manometric)

Recycled water pressure (around a border of the unit): $p_1 = 0.44 \text{ MPa}$ (manometric)

Atmosphere pressure for cooling tower: 1020 mBar (millibar)

Fouling resistance: $3.44 \times 10^{-4} \text{ m}^2 \cdot \text{k/v}$

Cycle of concentration: $N = 3.0$ (process water is used for feed water).

10.3.9.2.2. WATER DISCHARGE

Rainfalls

Table 10.3.9 – Average annual falls, mm.

January	February	March	April	May	June	July	August	September	October	November	December	Annual
14	12	13	14	16	19	16	12	12	13	13	18	172

Most falls per day, 58 mm

Snow mantle

Snowfall, snow depth and icing-up.

Snowfall is 65- 200 mm annually.

Table 10.3.10 – Snowfall

Snow mantle		
Averaged data of appearance and fracture of the stable snow mantle	Mean height for winter, cm	Maximum height for winter, cm
From December 10 to March 4	10	33
Icing-up effect		
Thickness of icing-up by a region	Standard icing-up with frequency in 5 years, mm	Standard icing-up with frequency in 10 years, mm
II	5	10

Tabular data is provided by Atyrau meteorological station, 2012.

Within the gas turbine power plant there is a network of storm sewage system for collection of rain and melt waters from roof of the power plant.

Pure surface runoff is discharged by the storm water sewage system to the reservoir – pure storm runoff ponds. Water from reservoir is pumped by submerged pumps to the pure surface water system of the complex.

Design water flow in the storm water sewage system at square inclined roof area of 16,570 m² is 176 l/s. Daily maximum is 980 m³.

The sewage system of a potentially contaminated oily runoff collects the surface runoff from road surfaces of the power plant to the contaminated surface runoff reservoir.

After primary cleaning, the runoff is discharged to the system of the complex.

The contaminated surface runoff reservoir collects a part of runoff more concentrated by contaminants (first portions of rain). The remaining runoff is discharged to the reservoir – pure storm runoff ponds. In this regard, at least 70% of annual volume of runoff is purified.

Design flow of surface waters in the system at area of passage cover of 9,150 m² is 150 l/s. Daily maximum is 550 m³.

Waste water sewage system of chemical water purification is designed for discharge of waste waters of the sample system, steam generator blowdown, waste waters of laboratory to the reservoir of chemically polluted waste waters. Water from the reservoir is pumped by submerged pumps to the neutralization system of the complex.

Oily waste water sewage system is designed for discharge of waste waters of the area of transformers, turbine house of steam turbines and steam generator, area of natural gas to the oil-content

waste water reservoir. Water from the reservoir is pumped by submerged pumps to the neutralization unit of the complex.

Domestic sewage system is designed for discharge of sanitary waste waters to the sanitary sewer system of the complex of the Integrated gas chemical facility.

10.3.10 OPERATION AND PRODUCTION CONTROL. LABOR AND WORKING CONDITION MANAGEMENT

This section demonstrates the solutions on production control, staff size, forms of labor management and layout of workspace, safety measures of the gas turbine power plant being part of the National Industrial Petrochemical Technological Park, located 47 km north-east to Atyrau.

Technical solutions of allied sections of the project have been accepted as benchmark data at developing this section: master plan and transport, process solutions, architectural & civil engineering solutions, fire prevention measures, heating, ventilation and conditioning, water supply and sewerage, power supply, etc.

The regulatory documents applicable in Kazakhstan have been used when developing the section.

Organizational structure of the GTPP control is accepted in accordance with the recommended structure of control developed by RSM company and taking into account the control structure of the Integrated gas chemical facility in a whole.

GTPP is designed for production of power and steam for consumers of the National Industrial Petrochemical Technological Park and is an organization department of the complex, therefore part of control and maintenance functions is transferred to the Department on Power Plants of the complex.

GTPP control organizational chart was developed on the ground of assumption that repairs and maintenance will be carried out by external resources.

GTPP control structure is as follows:

General management is performed by a director who directly subordinates: senior manager on operational control, senior manager on operation and maintenance, manager on occupational safety.

Staff size is determined on the basis of "Norms of size of production staff of power stations with gas turbine fixed plants with capacity of 12-150 mW" developed by "UES of Russia" RJSC, recommendations of a developer of main equipment and taking into account the control system of the whole complex.

GTPP production staff size will be 33 persons.

Amenities are designed at the first floor of the turbine house and consist of: cloakroom, shower rooms, wash rooms, toilets.

Near the cloakroom there are dining rooms, drinking facilities, portable air aggregates. Also, a staff room is provided near the switchboard.

The administrative and amenity building of the Integrated gas chemical facility has plant premises for dust removal and drying of work clothes and footwear, store rooms for overalls.

Plant canteen is located in the administrative and amenity building of the complex on the first floor. The canteen is designed as a food preparation station for semi-finished products and meant for provision of the staff of the whole complex with food.

General management of labor safety works is performed by the GTPP director. Line management of occupational safety service is performed by the HSE manager.

Meeting the safety requirements and creating the favorable working conditions for production staff is stipulated when designing the technological processes of the power station in strict adherence to the applicable norms, rules and determined by selection of a flow chart and layout solutions which provide the

staff with ease and safety of equipment operation, technological-process automation, use of protective means and facilities, maximum use of the mechanical aids when maintaining equipment.

Section 10.3.11 FIRE PREVENTION MEASURES

10.3.11.1 BACKGROUND INFORMATION

Section of the project on fire protection of GTPP buildings and structures, including: automatic systems of sprinkler, deluge, gas and dry powder fire extinguishing have been developed based on:

- design task;
- master plan;
- process solutions.

When designing, the following regulatory documents have been used:

SNiP RK 2.02-05-2009 “Fire protection of buildings and structures”

SNiP RK 2.02-15-2003 “Automatic fire fighting equipment of buildings and structures”

SN RK 2.02-11-2002 “Norms of providing the buildings, premises and structures with the systems of automatic fire alarm, automatic fire fighting plants and announcement of people about fire”.

Technical regulation dated 29.08.2008 No. 796 “Requirements to providing the buildings, premises and structures with the systems of automatic fire extinguishing and fire alarm, announcement and evacuation of people in case of a fire”.

RD 34.49.101-87 “Instruction on designing of fire protection of power companies”

“Industrial safety requirements for warehouses of liquified petroleum gases and pressure-operated highly inflammable liquids” were approved by the decree of the Minister for Emergency Situations of the Republic of Kazakhstan dated September 12, 2011 No. 380.

ST RK 1487-2006 “Fire engineering. Fire extinguishers. Requirements to operation”.

ISO 11602-1:2000 “Fire protection. Portable and transportable fire extinguishers. Part 1. Selection and installation”/

ISO 11602-2:2000 “Fire protection. Portable and transportable fire extinguishers. Part 2. Control and maintenance”.

10.3.11.2 Category of premises by fire and explosion and fire danger, fire resistance rating and classes of premises under the Electrical installation code of Kazakhstan were approved by the decree of the Government of the Republic of Kazakhstan dated October 24, 2012 No.1355.

Table 10.3.11 – Category of premises by fire and explosion and fire danger

No.	Building and premise name	Category of fire and explosion and fire danger under RNTF-01-94	Fire resistance rating of the buildings and structures (TP, annex 5)	Classification of premises and Electrical installation code
1	2	3	4	5
1	Turbine house	G	II	-
2	Voltage distribution cabinet	D	II	-
3	Cable floor	V	II	P- IIA
4	Switchboard	D	II	-
5	HVAC premises	D	II	P- IIA

6	Warehouse	D	II	P- IIA
7	Emergency diesel-generator room	G	II	P- I
8	Diesel fuel pipping house	V	II	P- I
9	Foam premise	D	II	-
10	Gas purification premise	A	II	V- IA
11	Premise for gas facilities	A	II	V- IA
12	Water supply premise	D	II	-
13	Battery room	D	II	V- IA
14	Condensate pump premise	D	II	-

10.3.11.3 LIST OF FIRE-FIGHTING SYSTEMS OF GAS TURBINE POWER PLANT

Table 10.3.11 – List of fire-fighting systems of gas turbine power plant

Premise, equipment	Fire extinguishing plant
Module of internal combustion turbine and accessories	Согласно стандарту производителя.
Turbine department	Indoor fire cocks. Indoor portable fire extinguishers.
Steam turbine, generator	Water fire-extinguishing system for bearings and oil module
Waste heat boiler	Fixed automatic spraying systems
Gas-collecting module	Outdoor fire hydrants
Electric power substation, premise for electrical equipment and control	IG541 or equivalent gas under the raised floor or computer-type floor, or indoor portable fire extinguishers.
Transformers	Indoor portable fire extinguishers
Diesel generator	As per manufacturer's standard
Ventilation systems	Smoke and fire damper
Diesel storage tank	Indoor foam extinguishing system and cooling of outdoor skeleton
Cylinder of diesel storage tank	Foam fire-fighting monitors
Diesel fuel pumps	Water nozzle

10.3.11.4 POWER STATION FIRE PROTECTION

GTPP fire-fighting system includes the following systems and equipment:

- Fire-fighting main.
- External pit fire hydrants.
- Indoor fire cocks.
- Fire protection by sprayed water for oil transformers.
- Fire protection by sprayed water for cooling of skeleton of the diesel storage tank.
- Fire protection by sprayed water for bearings of generator of internal combustion turbine and modules with lubricant oil for steam turbine.
- Foam system for diesel fuel tanks, dike for tank and pumps.
- Gas extinguishing system for hole under the raised floor.

- Gas extinguishing system for operator's room and cable ducts and basement.
- Portable and transportable fire extinguishers.

Fire-fighting system and fire protection equipment shall be supplied with the certificate of local GOST-K and fire permit of the State Technical Supervision Body of the Republic of Kazakhstan (Fire Safety Certificate of Kazakhstan).

The fire-fighting system for gas turbine and emergency diesel generator (set of equipment) are included by the manufacturer into the scope of this equipment.

10.3.11.5 FIRE-FIGHTING SYSTEMS OF GAS TURBINE POWER PLANT

10.3.11.5.1 FIRE-FIGHTING MAIN

For water supply of outer hoses, indoor cocks and fire-fighting units, an external fire water ring-main is designed which will be connected with external fire-fighting main of the whole complex and is included in the common fire-fighting system of the Integrated gas chemical facility.

To prevent freezing, the junction of underground system of fire water and fire protection equipment, and automatic drainage will be installed at depth of at least 0.5m below the frost level 1.7m.

10.3.11.5.2 DESIGN WATER FLOW FOR FIRE EXTINGUISHING

Main volume of fire water required for gas turbine power plant is as follows:

- 100 m³/h., given that the main transformer is on fire;
- 71 m³/h., given that the diesel fuel pump is on fire;
- 29 m³/h., given that the diesel storage tank is on fire (protection of internal foam system);
- 160 m³/h., given that skeleton of the diesel storage tank is on fire (fire protection by water spraying);
- 266 m³/h., given that the cylinder of the diesel storage tank is on fire (fire-fighting monitor).

The most disadvantaged fire dangerous conditions require the following maximum quantity of water for fire extinguishing:

- 266 m³/h., the diesel storage tank is on fire;
- 160 m³/h., cooling of the skeleton of the diesel storage tank;
- 108 m³/h., two fire hydrants (15 l/sec. each at pressure of 6 bar).

Total volume of the required fire water is 534 m³/h and is kept in fire tanks of the complex.

10.3.11.5.3 OUTDOOR FIRE HYDRANTS

Outdoor hydrants are designed per each 50m around the power station and not less than in 2.5m from the road side and 5m from buildings.

Outdoor fire hydrant shall be able to contain the following equipment for fire extinguishing:

- 2 fire hoses with Ø 65mm x 20 meters
- 2 spray equipment with float with Ø 65mm
- 2 hydrant spanners to secure the fire hose
- 1 name plate
- 1 adapter of stand pipe for pit hydrant under GOST with main valve and bivalve outlet.

10.3.11.5.4 INDOOR FIRE COCKS

Indoor fire cocks are installed inside the building in the turbine house. Inner fire cocks are connected to the GTPP external fire-fighting network.

Indoor hydrants shall be secured in the wall and/or column nearby the entrances and in any strategic location in the building of the turbine house. Each cabinet is equipped with galvanized and painted steel plate housing, glass door and has the following equipment:

- 1 absorption valve with diameter of 65mm;
- 1 flat fire hose with Ø 65mm x 20 m length;
- 1 spray equipment with float with diameter of 65mm;
- 1 spanner to secure the fire hose;
- 1 name plate.

10.3.11.5.5 FIRE-FIGHTING SPRAYING SYSTEMS OF TRANSFORMERS

Fixed fire-fighting spraying systems are designed for protection of the oil transformers with insulation. The fire-fighting spraying systems are activated by distribution control valves connected to the main fire-fighting network.

Filter is installed immediately after the distribution control valve to prevent contamination of the nozzles. Manual cut-off valve of indication is installed above each distribution control valve for reset and maintenance.

Distribution control valve is activated by fire detectors connected to the fire and gas detection and announcement system via solenoid valve.

Fire detectors are located in such order so as to activate the selection logic system.

Visual and sound units are provided to announce the staff about activation of the system.

Pressure relays are installed on the distribution control valves of the spraying fire-fighting system. Equipment for control of the distribution valves will be connected to the fire and gas detection and announcement system.

As per the Fire Regulations in Kazakhstan, Main Requirements PPB RK 08-97, the designed density of fire water for protection of transformers shall be 10.2 (l/min)/m^2 , for protection of the transformer surface and 6.1 (l/min)/m^2 for ground protection.

Distribution control valves for fixed water spraying system are installed on targeted module of stop valve with heating and ventilation (min. temperature is 5°C).

Hydraulic head in farthest water spraying nozzles shall be at least 1.5 bar.

10.3.11.5.6 FIRE EXTINGUISHING OF DIESEL STORAGE TANK

Fixed fire-fighting spraying systems are designed for protection of the skeleton of the diesel storage tank. The fire-fighting spraying systems are activated by distribution control valves connected to the main fire-fighting network. Fire-fighting spraying nozzles shall be of cone-type with medium speed. Filter is installed immediately after the distribution control valve to prevent contamination of the nozzles. Manual cut-off valve of indication is installed above each distribution control valve for reset and maintenance.

Distribution control valve is activated by fire detectors connected to the fire and gas detection and announcement system via solenoid valve.

Fire detectors are located in such order so as to activate the selection logic system. Remote activation of the distribution control valve is possible from local control room. Visual and sound units are provided to announce the staff about activation of the system.

Pressure relays are installed on the distribution control valves of the spraying fire-fighting system. Equipment for control of the distribution valves will be connected to the fire and gas detection and announcement system.

As per the Fire Regulations in Kazakhstan, Main Requirements PPB RK 08-97, the designed density of fire water for tank skeleton protection shall be 6.1 (l/min)/m^2 , fire water flow is $160 \text{ m}^3/\text{hour}$, 2675 l/min .

Distribution control valves for fixed water spraying system are installed on targeted module of stop valve with heating and ventilation (min. temperature is 5°C).

Hydraulic head in farthest water spraying nozzles shall be at least 1.5 bar.

10.3.11.5.7 FIRE EXTINGUISHING OF DIESEL FUEL PUMP

Fixed fire-fighting spraying systems are designed for protection of the diesel fuel pump. The fire-fighting spraying systems are activated by distribution control valves connected to the main fire-fighting network. Fire-fighting spraying nozzles shall be of cone-type with medium speed. Filter is installed immediately after the distribution control valve to prevent contamination of the nozzles. Manual cut-off valve of indication is installed above each distribution control valve for reset and maintenance.

Distribution control valve is activated by fire detectors connected to the fire and gas detection and announcement system via solenoid valve.

Fire detectors are located in such order so as to activate the selection logic system. Remote activation of the distribution control valve is possible from local control room. Visual and sound units are provided to announce the staff about activation of the system.

Pressure relays are installed on the distribution control valves of the spraying fire-fighting system. Equipment for control of the distribution valves will be connected to the fire and gas detection and announcement system.

As per the Fire Regulations in Kazakhstan, Main Requirements PPB RK 08-97, the designed density of fire water for tank skeleton protection shall be 20.4 (l/min)/m^2 , fire water flow is $71 \text{ m}^3/\text{hour}$, 1183 l/min .

Distribution control valves for fixed water spraying system are installed on foam firefighting station which will be heated and ventilated (min. temperature is 5^0C). Hydraulic head in farthest water spraying nozzles shall be at least 1.5 bar.

10.3.11.5.8 FOAM EXTINGUISHING SYSTEM

The following foam extinguishing equipment is designed:

- foam chamber of diesel storage tank with fixed cover;
- foam fire-fighting monitor for dike for diesel storage tank;

Aqueous film forming foam with 3% concentration will be a type of foam-blowing agent.

As per the Fire Regulations in Kazakhstan, Main Requirements PPB RK 08-97, the designed density of foamer solution for diesel storage tank and tank cylinder shall be:

- for surface of the tank with fixed cover: $4.1 \text{ (l/min.)}/\text{m}^2$ – 30 min. minimum supply time, foam flow is $29 \text{ m}^3/\text{hour}$, 479 l/min. ;
- area around the dike for tank (foam fire-fighting monitor): $6.5 \text{ (l/min.)}/\text{m}^2$ – 20 min. minimum supply time, foam flow is $266 \text{ m}^3/\text{hour}$, 4440 l/min. ;

Quantity of foam-blowing agent is calculated on maximum required designed density of supply of foamer solution in $\text{l/min.}/\text{m}^2$ to the zone of extinguishing for a period not less than the required minimum time of supply.

Capacity of the tank for foam-blowing agent for fixed system of foam extinguishing is calculated from the required capacity of foam-blowing agent plus 100% reserve. Fire extinguishing foam unit is fixed.

10.3.11.5.9 WATER EXTINGUISHING SYSTEM

High-pressure water extinguishing system is designed for the following sites:

- Oil module of steam turbine;
- Generator bearings of steam turbine;

The nozzles for water mist spraying produce a lot of small water drops resulting in more effective fire fighting by small quantity of water (usually 12 l/min per nozzle).

The water extinguishing system is designed for local application; for example, the water extinguishing system will be installed so as to supply water directly to the place of danger in protected or unprotected conditions. Drinking water is used for this system.

The water extinguishing system includes:

- Common base of module with:
 - A group of pressuring cylinders;
 - Pressure regulator valve / pressurizer valve;
 - Water body;
 - Pneumatic pumping unit;
 - Assembly of shutoff valves;
- Pipeline network;
- Nozzles.

The pneumatic pumping unit which is actuated by the group of pressuring cylinders supplies the required high pressure to the nozzles. Pump flow will be assessed in zones with maximum required water volume; for example, a zone which is protected by many nozzles.

Capacity of the groups of pressuring cylinders and water body will be enough for 30 minutes of application within the zone for which maximum water volume is required.

Valves of the group of pressuring cylinders and shutoff valves will be activated by the activation system. Water will be automatically supplied to the zone by thermal detectors which are connected to the system of fire and gas detection and announcement. Selecting logic system will be implemented for activation of water discharge. Visual/acoustic equipment will be provided for warning of staff about water discharge activation.

10.3.11.5.10 GAS EXTINGUISHING SYSTEM

Fire-smothering system is designed for premises of GTPP switchboard.

Water will be automatically discharged by smoke detectors which are connected to the system of fire and gas detection and announcement. Selecting logic system will be implemented for activation of water discharge. Visual/acoustic equipment will be provided for warning of staff about water discharge activation.

10.3.11.5.11 FIRE FIGHTING PRIMARY MEANS

Fire fighting primary means for GTPP buildings are accepted as per Annex 3, tables 1 and 2 of the “Fire Regulations in Kazakhstan”.

For arrangement of the fire fighting primary means, the fire racks are installed with a set of fire extinguishers, sand boxes, heavy sheets, bars, boat hooks, etc.

Man-handle and transportable (wheeled) fire extinguishers are located across the GTPP and buildings where they will be easily accessible in case of a fire.

Dry powder and wheeled chemical fire extinguishers used outdoors are designed for work at low external temperature.

Carbon dioxide portable and wheeled fire extinguishers are used indoors to fight a fire on electrical boards and equipment.

Dry powder and wheeled chemical fire extinguishers used indoors to fight a common fire.

Portable and wheeled foam fire extinguishers are used to fight an inflammation of mineral lubricating oil and diesel fuel. Portable fire extinguishers have 9 kg maximum capacity, and transportable fire extinguishers shall have capacity of at least 25kg.

10.3.11.6 AUTOMATIC FIRE-EXTINGUISHING AND FIRE ALARM SYSTEMS

Fire and gas detection system is required for first detection of fire, combustion gases and detection of intolerable concentration oxygen indoor.

As per SN RK 2.02-11-2002 “Standards of providing the building, premises and structures with the systems of automatic fire alarm, automatic fire-extinguishing and announcement of people about fire” (Annex B, table 2, Item 27), the power stations with G category as per fire danger shall be equipped with the announcement systems according to type III.

Fire and gas detection systems will automatically alarm, deal with fires or activate emergency shutdown (if required) to secure the staff and property.

Gas detection system has the following functions:

- acceptance of gas detector data in production premises and buildings;
- reception of gas alarm system and fault signal (repeated signals) from gas detector boards on the gas turbine;
- activates local gas detectors;
- displays information about gas alarm;
- actuates the emergency shutdown system of the heating, ventilation and conditioning system (if required);
- actuates the emergency shutdown system (if required);
- activates the emergency ventilation system;
- gives the repeated signals about gas alarm;
- gives the repeated signals about gas alarm at the Integrated gas chemical facility of the gas detection system;
- automatically controls all entrances/exits, field equipment and troubleshoot failure state.
- accepts data of fire detector in production premises and buildings for local priority control of sound signaling.

Fire, smoke, thermal and manual annunciators are located in premises and on production sites, taking into account arrangement of processing equipment.

As a device for centralized control of AFA system, the FGS controller is used which in a timely manner detects the attributes of burning, collects, processes, displays and registers the notifications about state of annunciators, controls a good condition of AFA loops, protects against faulty actuations.

FGS controller is installed in GTPP control room with 24x7 working staff and duplicated in switchboard control room.

FGS controller is supplied by category 1 from 220 V AC mains. For continuous and steady operation of AFA system, the specified equipment is completed by the uninterruptible power system.

AFA loop includes the targeted fire detectors:

Smoke;

Thermal;

Explosion-proof flame;

Manual and explosion-proof.

Automatic fire detectors shall be installed in different areas of GTPP fire danger, particularly:

Outdoor area:

flame detectors; thermal sensors; linear thermal sensors (heat-sensitive cable).

Turbine house: flame detectors; thermal sensors;

Control room buildings: high-sensitive smoke detectors (point laser); optical fire detectors; thermal sensors;

Electric substation buildings: radial smoke detectors.

Gas detectors shall be installed within an area of immediate possible source of leakage at a distance of 1m maximum. Gas detectors are installed on pumping units at a distance of 3m maximum from possible source of leakage.

AFA devices are mounted on the walls at height of between 1.7 and 2.4m from floor to operating controls.

Targeted detectors (fire, smoke and heat) are mounted on roof slabs (floor slabs) and on suspended ceiling. The targeted fire and explosion-proof detectors with informative signs (ST RK GOST R12.4.026-2002 "Signaling colors, safety signs and sign marking") are mounted on the walls of buildings and support at height of 1.5m from floor (ground).

Fire alarm and gas contamination alarm is provided both in a control room and within corresponding premises.

The power stations are equipped with the combustible gas detectors, taking into account the most probable sources of leakage, and taking into account their proximity to possible sources of ignition. Battery room shall be equipped with hydrogen detectors. Oxygen content detectors are installed in closed premises where possible leakage of inert substances can result in oxygen deficiency. Mark for alarm system response to oxygen deficiency is set at 19 %.

Combustible gas detectors detect two levels of low explosive limit: low level is for pre-alarm signaling and higher level is for alarm signaling. Level of alarm signaling will be registered and reset manually.

Level of pre-alarm signaling is usually set at 10% from low explosive limit, and level of alarm signaling is set at 25% from low explosive limit.

The manufacturer includes the fire and gas detection systems for gas turbines and emergency diesel generator in supply of this equipment.

To announce staff about fire, the main building premises and GTPP outside sites are equipped with the audible and light announcement lines which include the Light-audible annunciators.

Volume level of the alarm signal will be notably higher than noise level, however additional visual signal devices are provided for noisy places.

Fire alarms are used to announce about fire. Gas escape alarms are used for announce about gas escape.

Fire alarm sound is different from sound of the gas escape alarm or from any other alarm.

As a rule, in one acoustic zone the fire alarm prevails the gas escape alarm. When fire alarm is in operation, the gas alarm is off.

Fire bell is used to announce about fire inside buildings.

Stroboscopes and flashers are used as light flashing indicators which announce about fire or gas escape. They also add the warning horns. The different colours are used for announcing about highly flammable and toxic substances. Visual announcing shall be of rich colour which will be visible under any light.

Flashers are used as light flashing indicators which announce about fire or gas escape. Glasses of flashers shall be:

- RED is for announcing about fire;
- YELLOW/AMBER is for announcing about combustible gases;
- BLUE is for announcing about decrease in oxygen content.

Light-audible annunciators are mounted on the walls of building at height of at least 2.3m from the floor or ground and 0.1m from ceiling.

10.3.11.7 OCCUPATIONAL SAFETY

Automatic fire-fighting system shall be operated and maintained as required by the Fire Regulations in Kazakhstan.

Fire fighting plants shall be serviced by persons who were trained and attested as the specialists on plant maintenance and have a certificate of specialist.

Maintenance and duty staff shall know the operating principles of facilities and exercise the equipment regulations.

Equipment and industrial pipeline assembly and repair works are permitted only after check for absence of pressure in connection or pipeline.